

Drug-Discovery + Biomedical-Imaging ML — Research-Engineer Readiness — Master Learning Plan (v12 · DETAILED edition)

Owner: MD Hashim · **Start:** Sep 1, 2026 (Week 0 = bio-clock + new-project ramp; plan proper from Sep 8) · **Commitment:** ~40-42 h/week = 30 h structured weekday (Block A 06:00-08:00 · Block B 08:30-10:30 · Evening 20:30-22:30, x5) + ~10 h weekend buffer · **Total:** ~3,000 hrs · **Full-depth horizon:** ~mid-2028, laddered readiness across 2027 · **Targets:** ML Research Engineer / Applied Scientist at DeepMind-science, Isomorphic Labs, D. E. Shaw Research, Recursion/Insitro, top biomedical-imaging AI — and your organization's small-molecule & discovery-imaging AI work.

This is the DETAILED edition of v12. Every phase is elongated to the v2 pattern — week-by-week, with an **hour budget + how-to-use note + inline URL on every resource**, concrete **Deliverables**, **Whiteboard checkpoints**, per-phase **capstones**, and a phase **interview self-test**. Threads are expanded to the same depth, and the resource index is fused into one topic-organized list. Every squad-derived skill is rebuilt with **public methods + public datasets + personally-built capstones**; no internal codenames/targets/jira-IDs/teammate or partner names appear here or in any resume/public/interview artifact.

1. Weekly hours, timeline & a real-life feasibility check (read first)

	Hours	Content
Weekday structured (Mon-Fri)	30 h (6 h/day: Block A 2h + Block B 2h + Evening 2h)	new learning per the phase
Weekend buffer (Sat-Sun)	~10 h (2 x ~5 h)	revision, weekday spillover, emergencies — <i>not</i> new material by default
Total	~40-42 h/week	

Effort by block (cross-checked to ~2,925 h): Threads T+M+R ~550 (concurrent) · P0 55 · P1 210 · P2 core+Capstone1 220 · P2 imaging expansion 165 · P2V+Capstone2 100 · P3 165 · P4 165 · **Molecular-ML Track 5A-5F 450** · Phase Q+CapstoneQ 335 · P6+Capstone4 175 · P7+Capstone5 175 · P8 160.

Timeline (sequential spine + Phase Q at ~30 h/wk; threads run concurrently, so they add no calendar time): ~87 calendar weeks ≈ ~20 months → full depth ~mid-2028 — deliberately unhurried; see §5 for the week-by-week schedule. **Job-readiness is laddered far earlier** (apply from Week 3; Tier 2 by ~mid-Oct 2027; Tier 3 crescendo ~mid-Feb 2028).

Feasibility — honest verdict. 6 h of focused learning on top of a ~8.5 h office day is ~14-15 h of cognitive work/day — genuinely at the upper edge of sustainable. Guardrails are **load-bearing**: protect ~7 h sleep (lights-out ~22:45, wake ~05:45 — it's when the morning's learning consolidates); the **Evening block flexes first** when tired (push to weekend); one full rest day; consolidation weeks every ~5 weeks; plan 30 h, expect real delivery ~26-30 h and let the buffer absorb the gap; act on burnout signals early.

Parallel new project (critic agentic doc-review pipeline). Real GenAI/agentic work, directly on the Phase-7 path (critique/verify loops, MCP). Treat as office work, let it double as Phase-7 experience, keep its specifics off public artifacts, and don't let it eat Block A/B learning time.

2. The daily rhythm — cognitive-focus mapping

Window	State	Do this	Concretely
Block A · 06:00-08:00	Peak	Hardest NEW theory + derivation , intuition-first (visual → rigorous → derive by hand). Never email/video-binge.	New math; DL theory; GNN/GFlowNet/FEP/UQ/quantum theory.
Block B · 08:30-10:30	High, warmed-up	Active building & problem-solving. Turn Block-A theory into code.	Thread M (implement-from-scratch); capstone coding; Thread T DSA.
Evening · 20:30-22:30	Lower (post-work)	Review, intake, consolidation.	Lecture/seminar videos; paper reading (Thread R); Anki + glossary; re-do solved DSA; plan tomorrow's Block A.

Daily spacing loop: learn (A) → build (B) → review (Evening) → re-derive from memory next morning. **Weekends:** merge A+B into one ~4-h capstone deep-work session if caught up; keep evenings for review; protect one full day off.

3. What changed (v12 → v12-DETAILED)

- **Every phase (0-9, incl. the 5A-5F track) elongated to v2 depth:** week-by-week, hour-budgeted resources with inline URLs, deliverables, whiteboard checkpoints, per-phase capstones, and interview self-tests — self-contained, no cross-references.
- **Threads (§12) expanded** to the same depth.
- **Resource index fused** into one topic-organized list (§14) — the earlier v10-core / v11-additions split is gone.
- All hours, calendar, ladder, COI, and feasibility from v12 preserved.

4. Program at a glance

Block 1 — Foundations, imaging & molecular-ML core (Sep 2026 → mid-Oct 2027): P0 → P1 → **P2 imaging (expanded)** → P2V → P3 → P4 → **Molecular-ML Track (5A-5F)**. Tier 0 → Tier 2. **Phase Q — Quantum/QML** (parallel/after Tier-2): full depth; grounds 5D physics. **Block 2 — Scientific agents, CV breadth & interview crescendo** (mid-Oct 2027 → mid-Feb 2028): P6 → P7 → P8. Tier 3. **Threads (continuous from the Phase-1 self-test):** T (DSA+C++), M (implement-by-hand), R (research output). **Confusion Buffer** scaffolds every hard topic. **Application**

ladder runs from Week 3.

5. Master Calendar (sequential, from Sep 1, 2026, at ~30 h/wk structured; self-tests are the true gates)

Wee ks	Period	Phase (calendar weeks)	Milestone
0	Sep 1-7, 2026	bio-clock + new-project ramp; setup	schedule locked
1-3	Sep 2026	Phase 0 (2)	Phase-0 self-test → start applying (Tier 0)
3-13	late Sep – early Dec 2026	Phase 1 (10, incl. 2 consolidation)	Phase-1 self-test → Threads begin
13-27	Dec 2026 – mid-Mar 2027	Phase 2 core + expansion + Capstone 1 (14)	Tier 0/1 interview-ready
27-31	mid-Mar – early Apr 2027	Phase 2V + Capstone 2 (4)	Tier 1 (imaging + phenomics → Recursion/PathAI)
31-37	April – mid-May 2027	Phase 3 (6)	—
37-43	mid-May – late Jun 2027	Phase 4 (6)	—
43-58	late Jun – mid-Oct 2027	Molecular-ML Track 5A-5F (15)	Tier 2 — drug-discovery credible (DESRES/Isomorphic RE realistic)
58-64	mid-Oct – late Nov 2027	Phase 6 + Capstone 4 (6)	Tier 2 broadened
64-70	late Nov 2027 – early Jan 2028	Phase 7 + Capstone 5 (6)	—
70-76	Jan – mid-Feb 2028	Phase 8 interview crescendo (6)	Tier 3 loops — DeepMind / Isomorphic / DESRES. Intensive interviewing
76-87	~Feb – Apr 2028 (H1 2028)	Phase Q full + Capstone Q (11) — concentrated <i>after</i> the crescendo	QML branch opens; full depth ~mid-2028

Reading the calendar: Threads T/M/R and the application ladder run **concurrently inside the weekly hours** (not extra calendar time), which is why the ~2,925 total hours fit ~87 calendar weeks. **Phase Q is deliberately off the job-critical path** — concentrated after Phase 8 so it never delays Tier-3 interviewing; if a QML-specific role appears earlier, pull Phase Q forward and interleave. Consolidation weeks and the weekend buffer are already absorbed into the durations above.

Critical path to a DESRES/Isomorphic loop: P0 → P1 → P3 → **5A + 5D (+ conformer capstone) + 5E** → P4 system-design slice → P8, Thread M throughout. 5C, Phase Q, and Phase 7 deepen the profile but aren't on that path — so a credible Tier-2 loop is reachable around the **end of Phase 5 (~mid-Oct 2027)**, with the full crescendo by **~mid-Feb 2028**.

6. Application ladder

- **Tier 0 (from Wk 3):** India medical-imaging AI — Qure.ai [qure.ai](#) · SigTuple [sigtuple.com](#) · GE HealthCare [gehealthcare.com](#) · NVIDIA [nvidia.com](#) · MSR India [microsoft.com](#)
- **Tier 1 (+Capstone 2 + phenomics):** broader medical/biomedical imaging + video; **Recursion (phenomics — direct match)** [recursion.com](#) · PathAI · Paige · Owkin [owkin.com](#) · Google Research India [research.google](#)
- **Tier 2 (+Molecular-ML core + conformer/docking):** D. E. Shaw Research · Isomorphic Labs [isomorphiclabs.com](#) · Google DeepMind [deepmind.google](#) · Insitro [insitro.com](#) · Genesis [genesistherapeutics.ai](#) · Iambic [iambic.ai](#) · Chai [chaidiscovery.com](#) · InstaDeep [instadeep.com](#) · Valence [valencelabs.com](#)
- **Tier 3 (+interview crescendo):** full top-lab loops. **QML branch (+Capstone Q):** QpiAI [qpiai.tech](#) · IBM Quantum [ibm.com](#)
- *Your internal small-molecule / discovery-imaging directions are natural internal moves this public skill-set prepares you for — via normal channels, portfolio kept public.*

7. Risk register (compact)

Confusion compounding → the Confusion Buffer (understanding-gated self-tests, spaced re-derivation, triangulation). **Overload/burnout (the #1 risk at 40+ h/wk on top of work)** → sleep protection, weekend buffer, rest day, consolidation weeks; use them early. Breadth-without-depth → self-tests gate; Thread M enforces depth; the molecular track stays the priority. QML off the job-critical path. Everything portfolio-facing stays public and defensible.

8. Cross-cutting weekly habits

Paper-of-the-week (Evening) · Whiteboard-Friday concept (photo → prep deck) · Sunday teach-a-junior note · living glossary + war-chest. (Re-implementation → Thread M; reproduce-a-paper → Thread R.)

9. Publication & research-output ROI strategy

Publications are the write-ups of the capstones + monthly reproductions you're already doing — not extra projects (~1-2 weeks writing each). For an RE role a JD's "5+ publications" is a target, not the PhD-gated RS gate, and it counts workshop papers, preprints, **software papers (JOSS), reproducibility papers (ReScience/MLRC)**, and co-authorships — with first-authorship + open code + defensibility weighing more than raw count.

Highest-ROI moves (ranked): (1) each capstone → an arXiv/bioRxiv **preprint** (instant, citable) → a recognized **workshop** paper; (2) one **JOSS** software paper from a well-engineered capstone tool; (3) one **reproducibility** paper from Thread R; (4) a **co-authored** paper via collaborating with a lab whose work you reproduced; (5) a competition write-up (BraTS/Cell-Tracking/BELKA) or a focused benchmark.

Pipeline (over the plan's calendar):

Artifact	Output	Venue	~When
Capstone 1 (imaging seg)	preprint + workshop	MIDL / MICCAI-ws	preprint ~Mar 2027
Capstone 2 (video/lineage)	preprint + workshop	MIDL / CV-ws	~mid 2027
5A (equivariant-GNN benchmark)	preprint + workshop	MLSB / LoG	submit ~Sep 2027
5C (generative chemistry)	preprint + workshop	MLSB / LoG	~Sep 2027
5D (conformer/docking tool)	JOSS + workshop	JOSS + chem-ML ws	~Sep-Nov 2027
Thread R (reproduction)	reproducibility paper	ReScience / MLRC	~early 2028
Capstone 4 (scientific agent)	preprint + workshop	LLM-for-science ws	~late 2027
Collaboration	co-authored paper	lab's venue	2027-28

→ **~5-7 credible outputs by ~mid-2028**, mostly first-author, from existing work.

Guardrails: avoid predatory/pay-to-publish venues (they hurt credibility); prefer recognized workshops even if non-archival; **clear EVERY output through Lilly's external-publication/IP review before preprinting**; public data + generic methods only. **Venues:** MIDL [MIDL](#) · MICCAI [MICCAI](#) · MLSB [MLSB](#) · LoG [LoG](#) · JOSS [JOSS](#) · ReScience [ReScience](#) · arXiv [arXiv](#) · bioRxiv [bioRxiv](#) · OpenReview [OpenReview](#)

Full playbook: see the standalone **Publication_Strategy** document.

10. PhD-later decision guide (compact)

RS at frontier labs is PhD-gated; RE/Applied-Scientist generally isn't — RE is the near-term door, RS the post-PhD aspiration. **When:** after landing a strong RE role. **Where:** India external/part-time (IISc ERP [iisc.ac.in](#)), EU salaried/industrial (EURAXESS [euraxess.ec.europa.eu](#) — strongest funded route), US research-embedded. Set up now (zero cost): a capstone → workshop paper + preprint; 1-2 advisor relationships. Criteria: advisor fit → topic continuity → funding → part-time feasibility → location/visa → prestige (last).

11. Conflict-of-interest guardrails

Public data & generic methods only; personal hardware/accounts; **nothing proprietary or colleague-specific in any external artifact** (resume, public repos, interview talking points) — this includes the confidential squad briefings (their codenames/targets/Jira-IDs/names are for your skill-mapping only). Clear any public write-up through your org's external-publication review. Personal-hardware/public-data capstones aren't subject to the internal Artifactory rule; your Lilly work is.

12. Threads (continuous · ~550 hrs · start at the Phase-1 self-test, ~Week 10)

Do T & M in Block B (first ~45 min a thread, then phase coding); R in the Evening. Flex to ~2 hrs combined on consolidation/heavy weeks. These exist so no interview surface ever feels shaky.

Thread T — DSA + C++ · ~5 hrs/week · ~275 hrs

Stage T1 — C++ fundamentals (~40 hrs): types, references/pointers, RAII, STL containers/algorithms, classes, templates, move semantics, smart pointers.

- [learncpp.com](#) (sections 1-17, hands-on) — **~24 hrs** — [learncpp.com](#)
- The Cherno C++ (targeted: memory, pointers, move) — **~8 hrs** — [YouTube](#)
- [cpreference.com](#) (open while coding) — [en.cpreference.com](#) · Effective Modern C++ (Meyers) as reference · CppCon talks [YouTube](#) — **~8 hrs**

Stage T2 — DSA in C++, sustained (~235 hrs): ~4-6 problems/week, spaced then timed.

- NeetCode 150 (pattern-first) — [neetcode.io](#) · LeetCode — [leetcode.com](#)
- MIT 6.006 (algorithmic foundations) — [MIT OCW](#) · Abdul Bari (algorithms) [YouTube](#)
- Striver A2Z + SDE sheet — [takeuforward.org](#) · Sean Prashad patterns [seanprashad.com](#) · DesignGurus (Grokking patterns) [designgurus.io](#)
- CSES problem set + CP Handbook (for depth) — [cses.fi](#) · [cses.fi](#) **Checkpoint:** core structures + standard patterns from memory in C++; a random LeetCode **medium** in ~25-30 min, **hard** in ~40-45.

Thread M — Implement-by-hand ML · ~3 hrs/week · ~165 hrs · (highest ROI + concept-cementor)

From a **blank file, no autocomplete/AI**, one primitive in ~30-45 min, revisited spaced; add a weekly "build a small system" slot later. (Validated: DeepMind's RE coding round asks you to implement a loss from a blank file.)

- **Rotating primitives:** backprop through an MLP · scaled-dot-product + multi-head attention (+ $\sqrt{d_k}$) · Dice/Focal/Tversky · cross-entropy from logits · Layer/BatchNorm · stable log-softmax · a sampler (top-k/temperature) · a DDPM step · logistic/linear regression + gradients · k-means · PCA via SVD · SGD/Adam · positional encodings (sinusoidal/RoPE) · KV-cache · **a message-passing GNN layer** · **a small VQE circuit** · later **translate one to JAX**.
- References (read the mechanism, then close the tab): d2l.ai [d2l.ai](#) · labml.ai annotated [nn.labml.ai](#) · Annotated Transformer [nlp.seas.harvard.edu](#) · micrograd [github/micrograd](#) · nanoGPT [github/nanoGPT](#) · ML-From-Scratch [github/ML-From-Scratch](#) · minGPT [github/minGPT](#) · tinygrad [github/tinygrad](#) · JAX [jax.readthedocs.io](#) · Equinox [docs.kidger.site](#) **Checkpoint:** implement any listed primitive cold, timed, and explain each line.

Thread R — Research output, networking & applications · ~2 hrs/week · ~110 hrs

- **Reproduce-a-target-paper (monthly):** publicly reproduce one paper from a target team + write it up. — Papers with Code paperswithcode.com · How to Read a Paper (Keshav) web.stanford.edu · Connected Papers connectedpapers.com · Papers We Love [github/papers-we-love](https://github.com/papers-we-love)
- **Write:** 1-2 short posts/month; a "why me for digital biology" narrative; two job-talk decks.
- **Publish:** aim a capstone at a venue — MIDL midl.io · MICCAI miccai.org · MLSB mlsb.io · LoG logconference.org · arXiv arxiv.org · bioRxiv biorxiv.org · OpenReview openreview.net
- **Applications:** run the \$6 ladder; from Tier 0 this thread carries your live loops. **Checkpoint:** one reproduced paper + one write-up per month; a growing application pipeline.

13. THE PHASES (detailed)

PHASE 0 — Fundamentals Rebuild: Stats, ML & DL · ~55 hrs · Sep, Weeks 1-2 · → start applying (Tier 0)

Goal: rebuild crisp, out-loud command of the vocabulary and core ideas, intuition-first, building your living glossary as you go. By end you can explain ~15 fundamentals cold and you start applying to Tier-0 roles.

Week 1 — Statistics + classic-ML start (~28 hrs)

- **StatQuest — Statistics playlist** (distributions, variance/SD, CLT, sampling, hypothesis testing & p-values, CIs, R^2 , covariance/correlation, MLE, bootstrapping, power). Watch → write a glossary line + a tiny example for each. — ~14 hrs — [YouTube](https://www.youtube.com/watch?v=0j838318880) · statquest.org
- **StatQuest — classic ML (start)** (bias-variance, cross-validation, confusion matrix, precision/recall, ROC/AUC & when PR is honest). — ~9 hrs
- **Code drills** (Block B): simulate the CLT; a permutation test + bootstrap CI; a bias-variance curve. — ~5 hrs

Deliverable: the first ~40 glossary entries (your terminology war-chest v1). **Whiteboard checkpoint:** define a p-value and a 95% CI precisely; state the CLT; derive the MLE mean of a Gaussian; draw the bias-variance trade-off.

Week 2 — Classic ML + DL fundamentals + self-test (~27 hrs)

- **StatQuest — classic ML (finish):** linear/logistic regression; ridge vs lasso & **why L1→sparsity**; trees/RF/AdaBoost/GBM/XGBoost; SVMs & kernels; k-means/hierarchical; PCA; naive Bayes; entropy & mutual information. — ~11 hrs
- **DL fundamentals** — StatQuest NN + 3B1B "Neural Networks" ch 1-2 (neurons, layers, gradient descent, the shape of learning). — ~9 hrs — 3B1B NN [YouTube](https://www.youtube.com/watch?v=Vp08T0YnQj0)
- **Code drills:** logistic regression + L1/L2 from scratch (watch weights zero out); ROC & PR curves on an imbalanced toy set; a 1-neuron gradient-descent loop. — ~7 hrs

Deliverable (0.4): an articulation cheat-sheet (each concept in 3 plain sentences) + glossary v1. Reference: StatQuest *Illustrated Guide to ML*. **Phase 0 self-test (Feynman gate):** explain, out loud and with no notes — p-value, CI, CLT, MLE, bias-variance, ROC/AUC vs PR, why L1→sparsity, GBM vs RF, entropy/cross-entropy, gradient descent, over/underfitting. → **START APPLYING (Tier 0):** refresh CV/LinkedIn, first 3-5 quality applications.

PHASE 1 — Mathematics + Neural Networks from First Principles · ~210 hrs · Sep-Oct, Weeks 3-12 · a genuine build — do NOT rush

Goal: real mathematical maturity — the thing that breaks the vanilla-model ceiling and lets you explain *why* a model behaves as it does. Visual → rigorous → derive by hand, spaced. By end you can hand-derive backprop, implement micrograd from memory, and whiteboard attention with the $\sqrt{d_k}$ rationale — the gate that switches the Threads on.

Weeks 3-4 — Linear algebra (~72 hrs across the phase; front-loaded here)

- **3B1B — Essence of Linear Algebra** (build geometric intuition first; watch twice if needed) — ~8 hrs — [YouTube](https://www.youtube.com/watch?v=3B1B)
- **MIT 18.06 (Strang)** — selected lectures L1,3,5,6,9,10,14,15,16,21,22,25,29,30 + problems — ~30 hrs — [MIT OCW](https://ocw.mit.edu)
- **MIT 18.065** — SVD / PCA / low-rank / gradient-descent-through-a-LA-lens sections — ~18 hrs — [MIT OCW](https://ocw.mit.edu)
- Reference kept open: Matrix Cookbook math.uwaterloo.ca · (optional) Imperial Math-for-ML LA [Coursera](https://www.coursera.org)
- **Code drills (Block B):** vectors/matmul/transformations from scratch; eigendecomposition & SVD; truncated-SVD image compression; least-squares projection. — ~16 hrs

Deliverable: an LA notebook (SVD-PCA + low-rank compression + a projection) with a written "geometry of each operation" note. **Whiteboard checkpoint:** derive SVD from the eigendecomposition of $A^T A$; show why $A^T A$ is PSD; explain geometrically what a rank-r approximation does (Eckart-Young); derive the least-squares normal equations.

Week 5 — 🎯 CONSOLIDATION (~18 hrs, no new material)

Re-derive from memory (matmul-as-composition, determinant meaning, SVD-from-eigendecomposition, projection); re-implement SVD-PCA from a blank file; full Anki/glossary catch-up. **Do not skip.**

Weeks 6-7 — Calculus + matrix calculus (~26 hrs)

- **3B1B — Essence of Calculus** (derivatives, chain rule, the fundamental ideas) — ~6 hrs — [YouTube](https://www.youtube.com/watch?v=3B1B)
- **Matrix Calculus for Deep Learning (Parr & Howard)** — derive gradients of vector/matrix expressions — ~12 hrs — explained.ai · (optional) Imperial Multivariate Calc [Coursera](https://www.coursera.org)
- **Code drills:** gradient of a linear layer ($\partial L / \partial W = \delta x^T$) + numerical gradient check; softmax+cross-entropy forward/backward. — ~8 hrs

Deliverable: a from-scratch, gradient-checked linear layer + softmax-CE layer. **Whiteboard checkpoint:** derive backprop through a linear layer

with matrix calculus; derive $\partial(\text{softmax}+\text{CE})/\partial\text{logits} = p - y$; Jacobian vs Hessian.

Weeks 8-9 — Probability & statistics (~50 hrs)

- **Harvard Stat 110 (Blitzstein)** — targeted: probability, conditional probability & Bayes, random variables, expectation/variance, key distributions, conditional expectation — **~28 hrs** — projects.iq.harvard.edu · book (free) probabilitybook.net
- **MIT RES.6-012 (Intro to Probability)** — as a second teacher where Stat 110 is thin — **~12 hrs** — [MIT OCW](https://ocw.mit.edu)
- **Code drills:** simulate distributions; MLE fitting; a small Bayesian update; conditional-expectation-as-projection demo. — **~10 hrs**

Deliverable: a probability notebook (Bayes example + MLE fit + CLT simulation). **Whiteboard checkpoint:** Bayes' theorem with each term named; $E[X|Y]$ as an orthogonal projection; MLE for a Gaussian; why NLL minimization = MLE and cross-entropy = KL up to a constant.

Weeks 10-11 — Neural networks from first principles (Karpathy Zero-to-Hero) (~62 hrs) · PROTECT

- **Karpathy "Neural Networks: Zero to Hero"** — all 10 videos, **re-implement each without looking at his code** (micrograd → makemore incl. BatchNorm internals + manual backprop + WaveNet → GPT from scratch → tokenizer). — **~54 hrs** — [YouTube](https://www.youtube.com/watch?v=08G88383888) · hub karpathy.ai · repo [github/nn-zero-to-hero](https://github.com/karpathy/nn-zero-to-hero)
- **Thread M kick-off (Week 10):** reimplement backprop-through-MLP and multi-head attention from a blank file. — **~8 hrs**

Deliverable: micrograd + a small GPT, both built from scratch, in your `learning-log` repo. **Phase 1 self-test (gate → Threads T/M/R begin):** hand-derive backprop for a 2-layer MLP; implement micrograd from memory; explain why BatchNorm helps and what breaks without it; whiteboard scaled-dot-product attention and derive the $\sqrt{d_k}$ scaling.

Week 12 — 🎯 CONSOLIDATION (~18 hrs + light threads)

Re-derive backprop, attention+ $\sqrt{d_k}$, softmax-CE gradient, SVD from a blank page; re-implement micrograd-MLP and multi-head attention; Anki/glossary cleanup; a teach-a-junior note on attention. Threads continue light.

PHASE 2 — Applied CV & Biomedical Image Segmentation (EXPANDED) · ~195 hrs core (+ Capstone 1 ~25) + expansion ~165 hrs · Dec 2026 → mid-Mar 2027 · → Tier 0/1

Goal: own MRI / fluorescence / whole-slide segmentation end-to-end — physics → preprocessing → architectures → losses → 3D → uncertainty → evaluation — then extend into the discovery-imaging methods (phenomics, atlas registration, WSI engineering, pathology foundation models). By end you can whiteboard nnU-Net's design choices cold, defend each, and stand up a phenomics/WSI pipeline on public data.

Week 1 — Imaging physics (~30 hrs) — *most engineers skip this and it shows in their model design*

- **Allen Elster — Questions & Answers in MRI** (browse; the reference you'll return to) — **~8 hrs** — mriquestions.com
- **OHBM Educational — MRI physics** playlist — **~8 hrs** — [YouTube](https://www.youtube.com/playlist?list=PL8888888888888888)
- **Paul Callaghan — "Magnetic Resonance" (Otago)**, first 4 lectures — **~6 hrs** — [YouTube](https://www.youtube.com/watch?v=08G88383888)
- **iBiology — fluorescence microscopy primer** (your tissue work needs this) — **~8 hrs** — [YouTube](https://www.youtube.com/watch?v=08G88383888)

Deliverable: 1-page note — T1 vs T2 vs FLAIR vs T1CE (tissue appearance, why each exists, artifacts to preprocess: motion, bias field, partial volume); for fluorescence: photobleaching, channel bleed-through, autofluorescence and their effect on segmentation. **Whiteboard checkpoint:** which artifacts must be corrected before training and why; how MRI contrast maps to what a network can/can't learn.

Week 2 — CNNs, U-Net family, convolution math (~35 hrs)

- **Stanford CS231n** (focus L5, 9, 11 + notes) — **~14 hrs** — [YouTube](https://www.youtube.com/watch?v=08G88383888) · notes cs231n.github.io
- **Convolution arithmetic (Dumoulin & Visin)** — canonical for shapes/strides/padding/dilation, 2D & 3D — **~5 hrs** — [arXiv:1603.07285](https://arxiv.org/abs/1603.07285) · animations [github/conv_arithmetic](https://github.com/dumoulin/conv_arithmetic)
- **U-Net family — read in order, derive output sizes by hand** — **~16 hrs** — U-Net [arXiv:1505.04597](https://arxiv.org/abs/1505.04597) · V-Net (3D) [arXiv:1606.04797](https://arxiv.org/abs/1606.04797) · Attention U-Net [arXiv:1804.03999](https://arxiv.org/abs/1804.03999) · nnU-Net (read twice + supplementary) [Nature](https://www.nature.com)

Deliverable: a hand-computed layer-by-layer shape table for a U-Net on 256×256 input. **Whiteboard checkpoint:** draw U-Net from memory; compute output size at every layer; explain why skip connections work; explain why nnU-Net's "no new architecture" thesis is profound.

Week 3 — Loss functions for medical segmentation (your 95/5 imbalance) (~30 hrs) — *where segmentation pipelines die; own it cold*

- **Survey of segmentation losses (Jadon)** — derive each — **~6 hrs** — [arXiv:2006.14822](https://arxiv.org/abs/2006.14822)
- Generalized Dice (Sudre) [arXiv:1707.03237](https://arxiv.org/abs/1707.03237) · Focal Tversky [arXiv:1810.07842](https://arxiv.org/abs/1810.07842) · Boundary Loss [arXiv:1812.07032](https://arxiv.org/abs/1812.07032) — **~9 hrs**
- **From-scratch PyTorch (Thread M):** Dice, GDL, Focal, Tversky, Focal Tversky, Boundary, compound DiceCE; unit-test on synthetic imbalanced data; **plot gradients vs class ratio**; document when each helps. — **~15 hrs**

Deliverable: a tested losses library + a "gradient vs class-ratio" plot with a when-to-use guide. **Whiteboard checkpoint:** derive the Dice gradient and explain its instability on empty masks; why Focal/Tversky help severe imbalance.

Week 4 — nnU-Net deep dive + MONAI + microscopy (~45 hrs) — *the biggest hands-on block; nnU-Net stops being a black box*

- **MONAI tutorials** (clone, run, modify the brain-seg notebooks) — **~15 hrs** — [github/tutorials](https://github.com/monai/tutorials) · videos [YouTube](https://www.youtube.com/watch?v=08G88383888)
- **nnU-Net v2 — read the source** (teaches more than the paper) — **~8 hrs** — [github/nnUNet](https://github.com/nnUNet) · Kitware comparison [kitware.com](https://www.kitware.com)
- **DigitalSreeni (Python for Microscopists)** — the most microscopy-native channel; ~6-8 code-along videos mapped to your bottlenecks — **~14 hrs** — [YouTube](https://www.youtube.com/watch?v=08G88383888) · code [github/python_for_microscopists](https://github.com/python_for_microscopists)
- **AI for Medical Diagnosis (Rajpurkar)** — audit; the brain-MRI U-Net + evaluation modules — **~8 hrs** — [Coursera](https://www.coursera.org)

Deliverable: reproduce nnU-Net on a public brain dataset (BraTS subset / Decathlon task); write up which defaults it picked, why, and what you'd change for fluorescence microscopy. **Whiteboard checkpoint:** walk nnU-Net's fingerprinting → configuration; when it beats fancier architectures and when it doesn't.

Week 5 — Transformer-based segmentation + foundation models (~20 hrs)

- ViT (re-derive attention math) [arXiv:2010.11929](https://arxiv.org/abs/2010.11929) — ~4 hrs
- TransUNet / Swin-UNet / UNETR / Swin UNETR — ~10 hrs — [arXiv:2102.04306](https://arxiv.org/abs/2102.04306) · [arXiv:2105.05537](https://arxiv.org/abs/2105.05537) · [arXiv:2103.10504](https://arxiv.org/abs/2103.10504) · [arXiv:2201.01266](https://arxiv.org/abs/2201.01266)
- SAM / MedSAM [arXiv:2304.02643](https://arxiv.org/abs/2304.02643) · [Nature](https://www.nature.com/articles/s41586-023-04643-1) — ~6 hrs
- MedNeXt (a ConvNet that beats transformers on many medical tasks — keeps you honest) [arXiv:2303.09975](https://arxiv.org/abs/2303.09975) — read

Whiteboard checkpoint: inductive biases CNN vs ViT vs hybrid — when each wins; foundation models in low-data medical regimes; SAM's promptable design and what MedSAM changes.

Week 6 — 3D, uncertainty, evaluation + Capstone 1 (~35 hrs + Capstone 25)

- TorchIO (augmentation done right for volumes) — ~10 hrs — torchio.readthedocs.io
- Uncertainty (Yarin Gal — MC Dropout, deep ensembles, evidential) — ~8 hrs — [YouTube](https://www.youtube.com/watch?v=Uj8vYUg1o04)
- Metrics Reloaded (the definitive metric-choice paper) — ~5 hrs — [Nature](https://www.nature.com/articles/s41586-023-04643-1) · Domain shift (Stanford AIMI) aimi.stanford.edu
- 3D-conv whiteboard drill: input (D=64,H=128,W=128,C=4), kernel (3,3,3), stride (1,2,2), padding (1,1,1), 32 out — compute output shape + param count by hand; repeat for transposed & dilated. — ~3 hrs

CAPSTONE 1 (~25 hrs) · public data · Tier-0/1 gate: self-contained notebook + 3-page write-up on a public brain set (BraTS subset / Decathlon): (1) preprocessing with physics justification, (2) architecture ablation (3), (3) loss ablation, (4) uncertainty + failure-mode analysis; implement losses + one block from scratch. Datasets: Medical Decathlon medicaldecathlon.com · BraTS synapse.org · TCIA cancerimagingarchive.net · IXI brain-development.org · LIVECell [github/LIVECell](https://github.com/LIVECell) · BBBC bbbc.broadinstitute.org **Capstone-1 self-test:** whiteboard U-Net forward + skips with shapes; derive Dice gradient + empty-mask instability; why nnU-Net usually wins; for a new dataset, diagnose bias-field / imbalance / label-noise / domain-shift.

Imaging expansion modules (~165 hrs) — the discovery-imaging skill set (public data throughout)

2.6 — Cell/nucleus segmentation + phenomics (~35 hrs).

- Cellpose cellpose.org · StarDist [github/stardist](https://github.com/stardist) — ~12 hrs (run both on a public cell set).
- Cell Painting morphological profiling with CellProfiler; dose-response/EC50 + ROC-AUC validation — ~13 hrs — cellprofiler.org
- Datasets: BBBC bbbc.broadinstitute.org · JUMP-CP jump-cellpainting.broadinstitute.org · Recursion RxRx (public phenomics — direct Recursion-target match) rxrx.ai — ~10 hrs (build a small profiling pipeline). *Deliverable:* segment → extract morphological features → fit a dose-response/AUC readout on a public set. *Checkpoint:* what a Cell-Painting profile is; why EC50/AUC is the readout; batch effects in phenomics.

2.7 — 3D segmentation + atlas registration (~30 hrs).

- 3D sub-region segmentation + registration to a reference atlas — Allen Brain Atlas atlas.brain-map.org · BrainGlobe/brainreg brainreg.brainglobe.info · ANTsPy [github/ANTsPy](https://github.com/ANTsPy) · SimpleITK simpleitk.org — ~30 hrs. *Deliverable:* register a public mouse-brain volume to an atlas and quantify a per-region readout. *Checkpoint:* rigid vs affine vs deformable registration; what an atlas buys you; how registration error propagates to region metrics.

2.8 — Whole-slide / large-image engineering (~25 hrs).

- Tiling, pyramidal I/O, zarr/dask for 20-50 GB images; QC with QuPath/napari — OpenSlide openslide.org · TIAToolbox [github/tiatoolbox](https://github.com/tiatoolbox) · CLAM [github/CLAM](https://github.com/CLAM) · QuPath qupath.github.io · napari napari.org — ~25 hrs. *Deliverable:* a tiling + tiled-inference pipeline on a public WSI with stitched output. *Checkpoint:* why naive full-image loading fails; how to keep tile context; memory/throughput trade-offs.

2.9 — Stain normalization + MIL/weak-supervision + active learning (~25 hrs).

- Macenko/Reinhard/Vahadane normalization — torchstain [github/torchstain](https://github.com/torchstain) — ~8 hrs.
- Multiple-instance learning for weak WSI labels — CLAM [github/CLAM](https://github.com/CLAM) — ~10 hrs.
- Active learning to cut annotation — modAL [github/modAL](https://github.com/modAL) — ~7 hrs. *Deliverable:* an MIL slide-level classifier on a public WSI set with stain-norm ablation. *Checkpoint:* why stain variation breaks models; the MIL assumption; how active learning chooses samples.

2.10 — Pathology foundation models & SSL (~40 hrs) · strategic.

- DINOv2/MAE self-supervision — DINOv2 [github/dinov2](https://github.com/dinov2) · MAE [arXiv:2111.06377](https://arxiv.org/abs/2111.06377) — ~14 hrs.
- Use a pathology foundation model as feature extractor + fine-tune — UNI [github/UNI](https://github.com/UNI) · CONCH · Virchow (Nature Medicine 2024) [Hugging Face](https://huggingface.co/uni) — ~26 hrs (read the Virchow paper; run UNI embeddings + a linear/fine-tune head on a public task). *Deliverable:* compare supervised CNN features vs foundation-model embeddings on a public phenotypic/pathology task. *Checkpoint:* what SSL objective Virchow/DINOv2 use; why FM embeddings win in low-data; when fine-tuning beats linear-probe.

IMAGING-TRACK CAPSTONE (public): tile a large public WSI/microscopy image → segment → extract morphological/phenotypic features → fit a dose-response/AUC readout; compare CNN features vs foundation-model embeddings. One artifact showcasing segmentation + phenomics + foundation-model transfer — a MIDL/MICCAI-workshop candidate (Thread R).

PHASE 2 interview self-test: the Capstone-1 set **plus** — what a Cell-Painting profile encodes and its batch effects; registration types + error propagation; why WSI needs tiling/zarr and how to preserve context; the MIL assumption + stain-norm rationale; the SSL objective behind a pathology foundation model and when its embeddings beat a supervised CNN. → **Tier 0/1 (imaging + phenomics → Recursion/PathAI credible).**

PHASE 2V — Imaging & Video Extension (life-sciences-native) · ~70 hrs (+ Capstone 2 ~30) · mid-Mar - early Apr 2027 · → Tier 1

Goal: extend imaging into video/temporal — unlocking phenomics/cell-imaging over time and behavioral analysis. By end you can build a pose→behavior or cell-track→lineage pipeline and reason about tracking-by-detection vs joint methods.

Weeks 1-2 — Video & temporal models (~40 hrs)

- **Video transformers & motion** — VideoMAE [arXiv:2203.12602](#) · TimeSformer [arXiv:2102.05095](#) · ViViT [arXiv:2103.15691](#) · SlowFast [arXiv:1812.03982](#) — ~14 hrs
- **Segmentation/tracking over time** — SAM 2 (memory mechanism) [arXiv:2408.00714](#) · repo [github/sam2](#) · RAFT (optical flow) [arXiv:2003.12039](#) — ~12 hrs
- **Multi-object tracking** — ByteTrack [arXiv:2110.06864](#) · OC-SORT [arXiv:2203.14360](#) · BoT-SORT [arXiv:2206.14651](#) · HOTA metric [arXiv:2009.07736](#) — ~14 hrs — benchmarks: MOTChallenge [motchallenge.net](#) · DAVIS [davischallenge.org](#) · YouTube-VOS [youtube-vos.org](#)

Whiteboard checkpoint: SAM 2's memory mechanism; tracking-by-detection vs joint detection-tracking; when a video transformer beats frame-wise + tracking; the HOTA metric.

Weeks 3 — Life-sciences-native video (~30 hrs) — *the differentiator*

- **Pose estimation** — DeepLabCut [deeplabcut.org](#) (paper [Nature](#)) · SLEAP [sleap.ai](#) — ~14 hrs
- **Cell tracking & lineage** — Cellpose [cellpose.org](#) → TrackMate [imagej.net](#) / Ultrack [github/ultrack](#) — ~16 hrs — Cell Tracking Challenge [celltrackingchallenge.net](#)

CAPSTONE 2 (video) · public data · Tier 1: DeepLabCut/SLEAP pose → temporal model → behavior classification; **or** Cellpose seg → TrackMate/Ultrack tracking + lineage; SAM 2 for mask propagation. Datasets: CalMS21 [data.caltech.edu](#) · MABe [aicrowd.com](#) · Cell Tracking Challenge [celltrackingchallenge.net](#) **Phase 2V self-test:** SAM 2 memory; tracking-by-detection vs joint; when a video transformer wins; cell-lineage failure modes. → **Tier 1 unlocked (phenomics/behavioral imaging).**

PHASE 3 — Modern Deep Learning Theory + Foundation Architectures · ~165 hrs · *April - mid-May 2027* · → Tier 1 (broad)

Goal: stop treating transformers and diffusion as black boxes. Derive every layer of attention, understand optimization dynamics and modern training tricks, and defend architecture choices on theoretical grounds. This deepens skills you already use *and* is the prerequisite for Phase 4 and the generative parts of the molecular track (5C).

Week 1 — Optimization dynamics (~30 hrs)

- **Ruder — gradient-descent overview** (SGD → momentum → Adam family) — ~6 hrs — [ruder.io](#)
- **Boyd EE364A — Convex Optimization** (selected: convex sets/functions, gradient/Newton, duality intuition) — ~20 hrs — [YouTube](#) · book [web.stanford.edu](#)
- **Code drills:** implement SGD/Momentum/Adam from scratch; visualize on a 2D loss surface; a LR-warmup/schedule experiment. — ~4 hrs

Whiteboard checkpoint: why momentum accelerates; what Adam's second moment does; what warmup/schedules change in training dynamics; convex vs non-convex and why DL still works.

Week 2 — Information theory (~20 hrs)

- **MacKay — Information Theory, Inference & Learning (ch 1-6)** — entropy, KL, mutual information, coding — ~20 hrs — [inference.org.uk](#)

Whiteboard checkpoint: derive cross-entropy from KL; NLL minimization = MLE; where mutual information appears in contrastive losses (InfoNCE).

Week 3 — Transformers in depth + modern variants (~44 hrs)

- **Stanford CS25 — Transformers United** (6-8 talks) — ~16 hrs — [web.stanford.edu](#) · playlist [YouTube](#)
- **The Annotated Transformer** (code + math line by line) + **Lilian Weng** blog — ~16 hrs — [nlp.seas.harvard.edu](#) · [lilianweng.github.io](#)
- **Modern tricks (~30 min each):** RoPE [arXiv:2104.09864](#) · RMSNorm [arXiv:1910.07467](#) · FlashAttention [arXiv:2205.14135](#) · GQA/MQA [arXiv:2305.13245](#) · SwiGLU [arXiv:2002.05202](#) · MoE/Switch [arXiv:2101.03961](#) — ~12 hrs

Deliverable: a from-scratch transformer block (multi-head attention as one matmul, RoPE, RMSNorm, SwiGLU) with a param-count derivation.

Whiteboard checkpoint: derive scaled dot-product attention; multi-head as a single matmul; RoPE geometrically; why pre-LN beats post-LN; compute a block's param count from `d_model`, `n_heads`, FFN ratio.

Week 4 — Vision transformers + modern ConvNets (~27 hrs)

- **ViT lectures (Lucas Beyer)** — ~8 hrs — [YouTube](#)
- **ConvNeXt** [arXiv:2201.03545](#) · **DINOv2** (SSL, label-scarce) [arXiv:2304.07193](#) · **MAE** [arXiv:2111.06377](#) — ~12 hrs
- **HF Computer Vision course** (units new to you) — ~7 hrs — [Hugging Face](#)

Whiteboard checkpoint: ViT patch embedding; CNN vs ViT inductive biases; why SSL helps in low-data (ties to Phase 2.10).

Week 5 — Diffusion + generative modeling theory (~44 hrs) · PROTECT — feeds 5C

- **Stanford CS236 — Deep Generative Models** (L1-3, 6-8, 11-13) — ~22 hrs — [YouTube](#)
- **DDPM** [arXiv:2006.11239](#) · **Yang Song — score-based unification + blog** [yang-song.net](#) · **Lilian Weng diffusion** [lilianweng.github.io](#) — ~14 hrs
- **HF Diffusion course** (units 1-2, hands-on) — ~8 hrs — [Hugging Face](#)

Deliverable: a minimal DDPM trained on a toy set (implement the forward/reverse + the ϵ -loss from scratch — Thread M). **Phase 3 self-test:** implement attention on a whiteboard in 5 min; derive the diffusion loss (variational + score-matching views); why LayerNorm not BatchNorm in transformers; positional encodings compared (sinusoidal/learned/RoPE/ALiBi); compute a transformer block's param count.

PHASE 4 — Full-Stack / Production AI Systems + MLOps · ~165 hrs · *mid-May - late Jun 2027*

Goal: upgrade from "I built it" to "I can defend every architectural choice and design the next-gen version" across LLM systems, RAG, MLOps,

distributed training, and serving — and be fluent in ML system design (a core interview surface, and directly useful to your critic-agent project).

Week 1 — LLM internals (~34 hrs)

- **Stanford CS336 — Language Modeling from Scratch** (currently the best LLM-systems course) — ~26 hrs — [stanford-cs336.github.io](#) · playlist [YouTube](#)
- **HF NLP course** (units new to you) — ~8 hrs — [Hugging Face](#)

Whiteboard checkpoint: tokenization → embedding → blocks → head; KV-cache and why it matters; sampling (greedy/top-k/nucleus/temperature).

Week 2 — RAG at senior level (~28 hrs)

- **Hamel Husain — "What we learned from a year of building with LLMs" (I-II)** — ~8 hrs — [oreilly.com](#)
- **Eugene Yan — LLM patterns + Anthropic — contextual retrieval** — ~8 hrs — [eugeneyan.com](#) · [Anthropic](#)
- **Reranking, hybrid search, query rewriting, evals** — LlamaIndex [docs.llamaindex.ai](#) · RAGAS [docs.ragas.io](#) · DSPy [dspy.ai](#) — ~12 hrs

Deliverable: a RAG system with hybrid retrieval + reranking + a RAGAS eval harness on public docs. **Whiteboard checkpoint:** chunking/embedding/retrieval/rerank trade-offs; how you'd evaluate a RAG system honestly; where a critic/verifier agent improves it (ties to your project).

Week 3 — MLOps + ML systems design (~44 hrs) · KEEP

- **Chip Huyen — Designing ML Systems / interviews book** — ~18 hrs — [huyenchip.com](#)
- **Stanford CS329S — ML Systems Design** — ~14 hrs — [stanford-cs329s.github.io](#)
- **Made With ML** (hands-on) + **Full Stack Deep Learning** — ~12 hrs — [madewithml.com](#) · [fullstackdeeplearning.com](#)

Deliverable: a written ML-system-design doc for a real scenario (e.g., a molecular-property serving system, or a brain-seg pipeline ingesting new-institution data weekly with domain-shift handling). **Whiteboard checkpoint:** design a training→serving→monitoring loop with data/feature/model/serving layers; drift detection and retraining triggers.

Week 4 — Distributed training + inference optimization (~26 hrs)

- **Sebastian Raschka** (distributed training + efficient fine-tuning) — ~8 hrs — [magazine.sebastianraschka.com](#)
- **HF — efficient training on multiple GPUs** — ~6 hrs — [Hugging Face](#)
- **vLLM (paper + run it)** + quantization (GPTQ/AWQ/GGUF/FP7) + **ZeRO** — ~12 hrs — [arXiv:2309.06180](#) · ZeRO [arXiv:1910.02054](#)

Whiteboard checkpoint: data vs tensor vs pipeline parallelism; what ZeRO shards; how KV-cache + paged attention speed inference; the accuracy/latency of quantization.

Week 5 — Eval / observability / safety + systems-design prep (~33 hrs)

- **Anthropic — Constitutional AI + Eugene Yan — LLM evals** — ~10 hrs — [Anthropic](#) · [eugeneyan.com](#)
- **Observability** — run one of LangSmith / Phoenix / Helicone on a real RAG system — ~8 hrs
- **Systems-design interview prep** — whiteboard, recorded, self-critiqued: "RAG assistant for HCPs, 10K queries/day on 50K docs"; "brain-seg pipeline with weekly new-institution data + domain-shift handling"; "retraining + monitoring for a drug-discovery digital twin." — ~15 hrs — Chip Huyen book [huyenchip.com](#)

Phase 4 self-test: design an LLM-eval harness (incl. an LLM-as-judge with its pitfalls); a full ML-system-design whiteboard in 45 min; defend a serving/monitoring architecture; where a critic-agent adds value and how you'd evaluate *it*.

PHASE 5 · MOLECULAR-ML / DRUG-DISCOVERY TRACK (Phases 5A-5F)

Placement: Weeks 43-58 of v12 (~late Jun → mid-Oct 2027, **15 weeks**) · **~450 hrs** · **PRIORITY: high** — this is the differentiator that turns "strong imaging engineer" into "ML researcher for drug discovery," and the unlock for **Tier 2** (D. E. Shaw Research, Isomorphic Labs, DeepMind-science, Recursion/Insitro).

Goal of the whole track. By the end you can, on **public data**, cold and unaided: build and evaluate molecular property predictors across the full representation ladder (fingerprint → D-MPNN → 3D-equivariant); pretrain and finetune a molecular foundation model; generate diverse high-reward molecules with GFlowNets and equivariant diffusion; run and reason about the physics stack (conformer → docking → FEP → MD) and place every method on the accuracy/cost ladder; put **valid, calibrated uncertainty** on any prediction (conformer, GP/SVGP, ensembles); and ship the applied discovery methods a small-molecule squad actually uses (retrosynthesis, DEL, multitask, federated). Every claim here must be defensible in a DESRES-grade panel — the deliverables and whiteboard checkpoints exist to force that.

Prerequisites & sequencing. Needs Phase 1 (linear algebra, calculus/matrix-calculus, probability, backprop) and benefits from Phase 3 (GNN/attention/diffusion theory sits under 5A and 5C). **Runs alongside Phase Q** — the quantum-chemistry work grounds 5D's physics (why DFT/FEP are accurate and where they break). Do the sub-phases in order: **5A is foundational** (everything else assumes molecular representations); 5D is the heaviest and most DESRES-relevant; 5E is cross-cutting and can be pulled earlier if a UQ-heavy role appears.

How the hours land in your day (recap of the v12 rhythm). Block A 06:00-08:00 = the hard new theory/derivation of the day; Block B 08:30-10:30 = build it in code + the day's Thread-M/Thread-T work; Evening 20:30-22:30 = paper reading, re-derivation, Anki, planning. Weekends carry the capstone deep-work (uninterrupted blocks) and absorb spillover. At ~30 structured hrs/week, ~450 hrs ≈ **~15 weeks**.

Sub-phase summary

Sub-phase	Focus	Weeks (of track)	Approx. hrs
5A	Geometric & molecular ML core: GNNs → D-MPNN → 3D-equivariant	1-3	~90
5B	Molecular foundation models & self-supervised pretraining	4-5	~50
5C	Generative chemistry: GFlowNets + equivariant diffusion	6-8	~70
5D	Physics-based structure & simulation: conformer → docking → FEP → MD	9-12	~110

5E	Uncertainty quantification & probabilistic ML	13-14	~55
5F	Applied cheminformatics: retrosynthesis, DEL, multitask, federated, VS	15-16	~75

5A — Geometric & Molecular ML Core · ~90 hrs · PRIORITY: foundational

Everything downstream assumes you can turn a molecule into a good representation and reason about it. This is where that happens.

Goal: own molecular representation learning from fingerprints to message-passing to 3D-equivariant nets, and know **cold when each wins**. By end you can implement a message-passing layer from a blank file, train Chemprop and an equivariant GNN that beat a fingerprint baseline on a public ADMET benchmark, and defend every representation choice.

Week 1 — Graph ML foundations + molecular representations (~30 hrs)

Build the graph-learning spine and set the baseline every later model must beat.

- **Stanford CS224W (Leskovec)** — selected lectures: intro & node embeddings, the GNN lectures, message passing, and GNN expressiveness (Weisfeiler-Lehman). Watch → re-derive the message-passing update; do the associated Colabs. — **~14 hrs** — web.stanford.edu · playlist [YouTube](#)
- **PyG "Introduction by Example" + official Colabs** — build GCN / GraphSAGE / GAT on a toy graph task; understand batching of graphs. — **~6 hrs** — pytorch-geometric.readthedocs.io
- **Molecular representations primer** — fingerprints (ECFP/Morgan) vs physicochemical descriptors vs learned embeddings; RDKit getting-started + a fingerprint/descriptor tutorial. — **~5 hrs** — RDKit rdkit.org · DeepChem tutorials deepchem.io
- **Baseline build** — featurize a public ADMET task with Morgan fingerprints + RDKit descriptors; train XGBoost/RF with **scaffold splits** (not random — this is the honest split for molecules). — **~5 hrs** — Therapeutics Data Commons (ADMET benchmarks) tdcommons.ai · MoleculeNet moleculenet.org

Deliverable: a *baselines* notebook (fingerprint + descriptor models) on one TDC ADMET task with scaffold splits and proper metrics (RMSE/R² or AUROC) — this is the yardstick every model in 5A-5F must beat. **Whiteboard checkpoint:** write the message-passing update (aggregate → update) and prove permutation-invariance of the readout; state the Weisfeiler-Lehman bound on GNN expressiveness and what it means practically; contrast fingerprints vs learned embeddings (when does a fingerprint+GBM still win?).

Week 2 — Message-passing for molecules (D-MPNN / Chemprop) + graph transformers (~30 hrs)

The workhorse of production molecular property prediction.

- **Chemprop / D-MPNN** — read "Analyzing Learned Molecular Representations for Property Prediction" (Yang et al. 2019), then install and train Chemprop on your Week-1 task; ablate directed vs undirected messages and with/without RDKit-feature concatenation. — **~12 hrs** — [github/chemprop](https://github.com/chemprop)
- **Implement a message-passing GNN layer from scratch** (pure PyTorch or PyG MessagePassing) on a small molecular set — forward + backward, gradient-checked. This is a Thread-M anchor. — **~10 hrs**
- **Graph transformers** — read one of Graphormer / GraphGPS / TokenGT; understand how graph structure (centrality, spatial, edge encodings) is injected into attention and when a graph transformer beats an MPNN. — **~8 hrs** — Graphormer [arXiv:2106.05234](https://arxiv.org/abs/2106.05234) · GraphGPS [arXiv:2205.12454](https://arxiv.org/abs/2205.12454)

Deliverable: Chemprop **and** your from-scratch MPNN beating the Week-1 baseline on the TDC task; a one-page note on directed-MPNN design (why bond/edge-centered messages reduce "tottering") and when you'd reach for a graph transformer. **Whiteboard checkpoint:** derive the D-MPNN edge-message update and explain the tottering argument; explain how a graph transformer encodes structure into attention; state the over-smoothing problem in deep GNNs and two fixes.

Week 3 — 3D / equivariant molecular networks (~30 hrs)

Where geometry enters — and where the DESRES/quantum-chemistry world lives.

- **Geometric Deep Learning (Bronstein et al.)** — the symmetry/equivariance lectures; the "5 Gs" framing. Watch → be able to state invariance vs equivariance precisely. — **~8 hrs** — geometricdeeplearning.com · proto-book [arXiv:2104.13478](https://arxiv.org/abs/2104.13478)
- **Equivariant nets — read in order, derive the equivariance property each time:** SchNet (continuous-filter conv) → DimeNet++ (directional/angular) → EGNN (simple E(n)-equivariance) → NequIP/MACE (tensor-product message passing for potentials). — **~14 hrs** — SchNet [arXiv:1706.08566](https://arxiv.org/abs/1706.08566) · DimeNet++ [arXiv:2011.14115](https://arxiv.org/abs/2011.14115) · EGNN [arXiv:2102.09844](https://arxiv.org/abs/2102.09844) · NequIP [Nature](https://www.nature.com/articles/s41586-021-03608-0) · MACE [arXiv:2206.07697](https://arxiv.org/abs/2206.07697)
- **e3nn tutorial + run an equivariant model on QM9** — train SchNet/EGNN-style on a QM9 property; verify equivariance empirically (rotate input → output transforms correctly). — **~8 hrs** — e3nn e3nn.org · tutorial blondegeek.github.io · QM9 quantum-machine.org

Deliverable: an E(3)-equivariant GNN predicting a QM9 property; ablate equivariance against a non-equivariant baseline and report the **data-efficiency** gain (accuracy at 1k / 10k / full labels). **Whiteboard checkpoint:** define invariance vs equivariance formally ($f(g \cdot x) = g \cdot f(x)$); show why 3D-equivariance matters for energies/forces; explain how EGNN updates coordinates equivariantly without spherical harmonics; state when a 2D-graph model beats a 3D one (and vice versa).

5A CAPSTONE (folded into the weeks, ~10 hrs of the 90) — "the representation ladder": on one public ADMET/QM9 task, compare **fingerprint+XGBoost vs D-MPNN (Chemprop) vs 3D-equivariant GNN** on the same scaffold split; report accuracy, data-efficiency, and compute cost; include your from-scratch message-passing implementation. Public repo + short write-up. *This single artifact is the strongest proof of molecular-ML competence you can show a panel.*

5B — Molecular Foundation Models & Self-Supervised Pretraining · ~50 hrs · PRIORITY: medium-high

Low-data is the norm in drug discovery; pretraining is how you win it. Also directly on the small-molecule squad's foundation-model thrust.

Goal: pretrain-then-finetune molecular models and know when SSL pays off. By end you can run a small masked-SMILES/graph pretrain, finetune it, and quantify the transfer gain vs training from scratch.

Week 4 — SSL objectives + SMILES/graph pretraining (~25 hrs)

- **Read the landscape (SSL for molecules):** ChemBERTa (masked-SMILES BERT), Grover (graph SSL), MolCLR (graph contrastive), Uni-Mol (3D-aware). Read → tabulate objective, representation, and what each pretraining signal captures. — **~10 hrs** — ChemBERTa [arXiv:2010.09885](#) · MolCLR [github/MolCLR](#) · Uni-Mol [github/Uni-Mol](#) · MolFormer [github/molformer](#)
- **Run a small masked-SMILES pretrain** on a public unlabeled set (e.g., a ZINC/ChEMBL SMILES subset) with a HF encoder — masking, tokenization, the MLM objective. — **~15 hrs** — HF course (for the mechanics) [Hugging Face](#) · ChEMBL [ebi.ac.uk](#) · ZINC [zinc.docking.org](#)

Deliverable: a working masked-SMILES pretraining run (loss curve + a sanity-check on masked-token recovery). **Whiteboard checkpoint:** write the masked-language-modeling objective; explain what a *valid* molecular augmentation is for contrastive SSL (and why bad augmentations break it); state the difference between what a SMILES-LM vs a 3D-SSL model learns.

Week 5 — Finetune + transfer study (~25 hrs)

- **Finetune a pretrained molecular encoder** (your Week-4 model, or a public checkpoint) on a small TDC task; compare vs your 5A from-scratch model. — **~15 hrs** — TDC [tdcommons.ai](#)
- **Transfer/data-ablation study** — measure the transfer gain at 100 / 1k / 10k labels; identify where pretraining helps vs where it's neutral. — **~10 hrs**

Deliverable (5B capstone): a pretrain→finetune notebook showing the transfer gain vs from-scratch across label-budget regimes, with an honest "when it didn't help" section. **Whiteboard checkpoint:** when does SSL pretraining help vs hurt a downstream molecular task? Why can a fingerprint+GBM still beat a fine-tuned foundation model on a small, well-defined endpoint?

5C — Generative Chemistry: GFlowNets + Equivariant Diffusion · ~70 hrs · PRIORITY: high (closes the "generative molecules" gap)

This is the DESRES wish-list item your resume can't yet claim. Build it and it becomes an honest, defensible bullet.

Goal: generate diverse, high-reward, valid molecules; understand GFlowNets (flow-matching) vs RL vs VAE/flow vs diffusion, cold; run a property-reward generator and evaluate it properly.

Week 6 — Generative baselines + the landscape + metrics (~25 hrs)

- **VAE/flow baselines** — skim JT-VAE and MoFlow; understand latent-space generation and its failure modes (validity, posterior collapse). — **~8 hrs** — JT-VAE [arXiv:1802.04364](#) · MoFlow [arXiv:2006.10137](#)
- **The generative-model landscape for molecules** — a survey pass: search-based vs autoregressive vs latent vs diffusion vs GFlowNet; the axes that matter (diversity, validity, novelty, optimizability). — **~7 hrs** — DeepChem generative tutorials [deepchem.io](#)
- **Metrics + reward design** — validity, uniqueness, novelty, diversity (internal distance), and property rewards (QED, SA-score, a docking/predicted-activity proxy); GuacaMol/MOSEs benchmarks. — **~10 hrs** — MOSES [github/moses](#) · GuacaMol [github/guacamol](#)

Deliverable: a metrics + reward module (validity/novelty/diversity + a property reward) reusable by the generators below. **Whiteboard checkpoint:** define validity/uniqueness/novelty/diversity precisely; explain why "high reward" alone is a bad objective (mode collapse) and why diversity matters in hit-finding.

Week 7 — GFlowNets (~25 hrs)

- **GFlowNet theory** — Bengio et al. "Flow Network based Generative Models" + "GFlowNet Foundations"; understand the flow-matching / trajectory-balance objective and the sample-diversity- α -reward guarantee. — **~12 hrs** — [arXiv:2106.04399](#) · foundations [arXiv:2111.09266](#)
- **GFlowNet hands-on** — work the tutorial/code; train a GFlowNet on a small fragment- or atom-based molecule-building environment with your Week-6 property reward. — **~13 hrs** — [github/gflownet](#)

Deliverable: a GFlowNet molecule generator with a property reward; report diversity/validity/novelty/reward vs a naive RL or random baseline. **Whiteboard checkpoint:** write the trajectory-balance (or flow-matching) objective; explain **why a GFlowNet samples proportionally to reward and therefore stays diverse**, whereas reward-maximizing RL collapses to a few modes; explain the DAG-of-states construction for molecule building.

Week 8 — Equivariant diffusion for 3D molecule generation (~20 hrs)

- **EDM (equivariant diffusion) + GeoDiff** — read; understand denoising over 3D coordinates with E(3)-equivariance; connect to torsional diffusion (conformers, from 5D). — **~12 hrs** — EDM [github/e3_diffusion_for_molecules](#) · GeoDiff [github/GeoDiff](#)
- **Run a small 3D generation** (pretrained or tiny-train) and evaluate geometry validity. — **~8 hrs**

Deliverable (5C capstone): the GFlowNet (or diffusion) generator + evaluation, as a public repo; a note comparing GFlowNet vs RL vs diffusion trade-offs for molecular design. **Whiteboard checkpoint:** how does equivariant diffusion denoise 3D coordinates while respecting rotation/translation symmetry? When would you choose a 2D-graph GFlowNet over a 3D diffusion model for de-novo design?

5D — Physics-Based Structure & Simulation · ~110 hrs · PRIORITY: high (the DESRES core; bridges Phase Q)

The "more accurate approaches for molecular simulation" leg. Owning the accuracy/cost ladder is what a computational-chemistry panel probes hardest.

Goal: run and reason about conformer generation, docking, free-energy, and MD, and place every method on the ladder MMFF → semi-empirical (GFN2-xTB) → ML-potential → DFT → FEP → MD. Reuses your DFT/quantum-chemistry strength.

Week 9 — Conformer generation & 3D structural analysis (~30 hrs)

Follow the standalone spec end-to-end (it is written for exactly this slot).

- **Conformer pipeline** — RDKit ETKDGv3 + MMFF94s optimization + Butina RMSD clustering; **COV/MAT/RMSD** benchmarking against **GEOM-Drugs** reference ensembles; **DFT re-ranking** of top conformers (Psi4/PySCF, ω B97X-D/def2-SVP) reusing your quantum-chem skill. — **~30 hrs** — RDKit rdkit.org · GEOM [github/geom](https://github.com/geom) · Psi4/PySCF · (ML potentials MACE-OFF/TorchANI for the accuracy-vs-cost curve)

Deliverable: the conformer-generation + 3D-analysis repo (COV/MAT/RMSD + energy-ranking correlation MMFF-vs-DFT). **Whiteboard checkpoint:** why is MMFF energy ordering unreliable and what fixes it? define COV vs MAT (recall vs precision); why symmetry-corrected best-RMSD; why def2-SVP + a dispersion correction.

Week 10 — Molecular docking (~25 hrs)

- **Classical docking** — AutoDock Vina / smina: search + scoring; prepare a public target (from PDBbind or DUD-E), dock a ligand set, evaluate pose RMSD and screening enrichment (EF/BEDROC). — **~13 hrs** — Vina vina.scripps.edu · PDBbind pdbind-plus.org.cn · DUD-E dude.docking.org
- **ML docking / scoring** — glna (CNN scoring) and DiffDock (generative pose prediction); compare to classical docking on the same target. — **~12 hrs** — glna [github/glna](https://github.com/glna) · DiffDock [github/DiffDock](https://github.com/DiffDock)

Deliverable: a docking mini-study on a public target — classical vs ML docking, pose RMSD + enrichment. **Whiteboard checkpoint:** what does a docking score approximate and why is it a poor absolute affinity? why is pose RMSD not enough without enrichment? what does DiffDock change vs search-based docking?

Week 11 — Free energy (FEP / ABFEP) (~25 hrs)

- **Statistical-mechanics intuition** — free energy, ensembles, thermodynamic cycles, why relative FEP cancels errors; alchemistry primer. — **~12 hrs** — alchemistry.org
- **Hands-on relative FEP** — OpenFE: set up a small relative binding free-energy calculation between two congeneric ligands; read the result critically. — **~13 hrs** — OpenFE openfe.energy

Deliverable: a documented relative-FEP example + a one-page "when is FEP worth the cost" note. **Whiteboard checkpoint:** draw the thermodynamic cycle behind relative binding FEP and explain what cancels; why FEP is accurate but expensive; where it fails (force-field quality, sampling, large perturbations).

Week 12 — Molecular dynamics (~30 hrs)

- **OpenMM basics** — force fields, integrators, thermostats/barostats, periodic boundaries; run a short protein-ligand MD; the "Making it Rain" colabs are the fastest on-ramp. — **~18 hrs** — OpenMM openmm.org · Making-it-Rain [github/Making-it-rain](https://github.com/Making-it-rain)
- **Trajectory analysis + ML potentials** — MDAnalysis (RMSD/RMSF/contacts); ML interatomic potentials (MACE-OFF / ANI-2x) as fast DFT-quality surrogates and their transferability limits. — **~12 hrs** — MDAnalysis mdanalysis.org

Deliverable (5D CAPSTONE): a conformer→docking pipeline on a public target (pose/enrichment) + a short OpenMM MD run with trajectory analysis + the FEP exercise; a write-up placing **every method on the accuracy/cost ladder** with your quantum-chem work at the top.

Whiteboard checkpoint: the full accuracy/cost ladder and where each rung breaks; what a force field is and its limits; why MD needs thermostats/barostats; how an ML potential reaches DFT-quality at MM cost and where it fails; how this connects to your VQE/DMET quantum work.

5E — Uncertainty Quantification & Probabilistic ML · ~55 hrs · PRIORITY: high (cross-cutting; a stated squad priority)

A prediction without calibrated uncertainty is not decision-grade. This is also a direct squad deliverable and a strong interview differentiator.

Goal: put valid, calibrated uncertainty on molecular predictions; know conformal prediction, GPs/SVGP, calibration, and ensembles/MC-dropout cold, and when to use each.

Week 13 — Conformal prediction + calibration (~28 hrs)

- **Conformal prediction** — Angelopoulos & Bates "A Gentle Introduction to Conformal Prediction"; split/inductive conformal; how it gives distribution-free marginal coverage. — **~14 hrs** — [arXiv:2107.07511](https://arxiv.org/abs/2107.07511)
- **MAPIE hands-on** — add conformal prediction intervals to your 5A property model on a public ADMET set; verify empirical coverage at target α . — **~8 hrs** — [github/MAPIE](https://github.com/MAPIE)
- **Calibration** — ECE, reliability diagrams, temperature scaling; what calibration does and doesn't tell you. — **~6 hrs**

Deliverable: conformal intervals on your 5A model with an empirical-coverage plot (does 90% nominal give ~90% coverage?). **Whiteboard checkpoint:** why split conformal gives valid marginal coverage without distributional assumptions; the exchangeability assumption and how scaffold/temporal shift breaks it; what ECE measures and its failure modes.

Week 14 — Gaussian processes / SVGP + the Bayesian view (~27 hrs)

- **GP regression** — kernels, the posterior mean/variance, what the kernel encodes; the intuition before the algebra. — **~10 hrs** — Murphy *Probabilistic ML* probml.github.io · (Rasmussen & Williams GPML as reference)
- **Sparse variational GP (SVGP) hands-on** — GPyTorch: an SVGP on a public ADMET set for scalable calibrated regression. — **~10 hrs** — GPyTorch gpytorch.ai
- **Compare the UQ toolbox** — conformal vs GP vs MC-dropout vs deep ensembles on the same task; honest trade-offs. — **~7 hrs**

Deliverable (5E CAPSTONE): add **both** conformal intervals and an **SVGP** to your 5A model; report empirical coverage + calibration; a four-way UQ comparison (conformal / GP / MC-dropout / deep-ensemble) with a recommendation. **Whiteboard checkpoint:** the GP posterior mean/variance formulas and what the kernel encodes; why deep ensembles are often the strongest practical UQ; conformal vs Bayesian coverage guarantees; when you'd ship which method.

5F — Applied Cheminformatics & Discovery ML · ~75 hrs · PRIORITY: medium-high

The applied methods a small-molecule squad ships. Pick a lane for the capstone based on the role you're closest to landing.

Goal: own retrosynthesis/reaction-condition, DEL modeling, multitask/MMoE, contrastive representation, federated learning, and active-learning virtual screening well enough to contribute and to defend in a panel.

Week 15 — Retrosynthesis + reaction-condition (~25 hrs)

- **Retrosynthesis (AiZynthFinder)** — MCTS over a building-block stock with a template/expansion policy; run it on public targets; understand the search. — ~13 hrs — [github/aizynthfinder](https://github.com/aizynthfinder/aizynthfinder)
- **Reaction representation + condition prediction** — reaction fingerprints/graphs; predict conditions (reagent/solvent/temp) on public reaction data (USPTO / Open Reaction Database). — ~12 hrs — Open Reaction Database open-reaction-database.org

Deliverable: a retrosynthesis run + a small reaction-condition model with honest top-k accuracy. **Whiteboard checkpoint:** how MCTS retrosynthesis searches (selection/expansion/rollout/backup); why synthesizability matters as a generative constraint; why reaction-condition prediction is multi-label and noisy.

Week 16 — DEL, multitask/contrastive, federated + virtual screening (~50 hrs)

- **DEL (DNA-encoded library) modeling** — enrichment/denoising from noisy barcoded counts; the public **BELKA** dataset. — ~15 hrs — kaggle.com
- **Multitask / MMoE + contrastive** — a multi-endpoint ADMET model (shared trunk, per-task heads); Mixture-of-Experts to reduce negative transfer; contrastive representation of binding/affinity. — ~15 hrs — (build on your 5A stack + TDC multi-endpoint tasks tdcommons.ai)
- **Federated learning + active-learning virtual screening** — Flower: a federated multi-endpoint ADMET demo (train across simulated sites without pooling data); an active-learning loop that docks + trains an ML surrogate over a library (ties 5A + 5D + 5E). — ~20 hrs — Flower flower.ai

Deliverable (5F CAPSTONE — pick ONE lane, ship a public repo + write-up):

1. **Agentic decision-support:** wrap your 5A property + 5E UQ model as an **MCP tool** with a synthesizability check — a "drug-hunter co-pilot" (ties to Phase 6 and mirrors your critic-agent doc-review project's patterns).
2. **Reaction/retrosynthesis pipeline** (Week-15 work, hardened).
3. **Federated multi-endpoint ADMET** demo.
4. **DEL enrichment** model on BELKA. **Whiteboard checkpoint:** why DEL counts are noisy and how enrichment modeling denoises them; when multitask helps vs negative transfer and how MMoE mitigates it; what federated learning trades off (privacy vs communication/heterogeneity); how active learning chooses the next compounds to screen.

PHASE 5 — TRACK INTERVIEW SELF-TEST (must pass cold, no notes → Tier 2)

Answer each out loud, whiteboard where relevant:

1. **Representations:** message passing vs convolution; what a D-MPNN buys over a fingerprint and when the fingerprint+GBM still wins; the WL expressiveness bound; over-smoothing and fixes.
2. **3D/equivariance:** invariance vs equivariance formally; why 3D-equivariance matters for energies/forces; how EGNN updates coordinates equivariantly; when 2D beats 3D.
3. **Pretraining:** the MLM objective; a valid molecular augmentation for contrastive SSL; when SSL helps vs hurts.
4. **Generative:** the GFlowNet objective and **why it yields reward-proportional diversity** (vs RL mode-collapse); how equivariant diffusion respects symmetry; validity/novelty/diversity metrics.
5. **Physics:** the full MMFF→xTB→ML-potential→DFT→FEP→MD **accuracy/cost ladder** and where each breaks; the thermodynamic cycle behind relative FEP; what a docking score really approximates; how your VQE/DMET work sits at the top of the ladder.
6. **Uncertainty:** why split conformal gives valid coverage without distributional assumptions and how molecular distribution-shift breaks exchangeability; GP posterior + what the kernel encodes; why deep ensembles are often best in practice; what ECE measures.
7. **Applied:** MCTS retrosynthesis; DEL count denoising; multitask vs negative transfer (MMoE); federated-learning trade-offs; the active-learning virtual-screening loop.
8. **System design (ties Phase 4/8):** design a molecular-property prediction service with calibrated UQ and scaffold-aware validation; design an active-learning virtual-screening pipeline over a billion-compound library.

Flagship artifacts this phase produces (for the resume/portfolio, all public): (5A) the representation-ladder study; (5C) a GFlowNet/diffusion generator; (5D) the conformer→docking→FEP→MD physics repo; (5E) the four-way UQ comparison; (5F) one applied capstone. Any one of these is a legitimate MLSB / LoG / AI4Science workshop submission — aim at least one there (Thread R).

PHASE 6 — Generative & Agentic AI (scientific / chemistry agents) · ~155 hrs (+ Capstone 4) · mid-Oct - late Nov 2027

Goal: own modern LLM adaptation, multimodal, and agentic systems — reframed toward science. By end you can wrap a model as an MCP tool, build a multi-step research/critique agent, and evaluate agents rigorously (directly reinforcing your critic-agent doc-review project).

Week 1-2 — Advanced LLMs / adaptation (~42 hrs)

- LoRA [arXiv:2106.09685](https://arxiv.org/abs/2106.09685) · QLoRA [arXiv:2305.14314](https://arxiv.org/abs/2305.14314) · Raschka (fine-tuning) magazine.sebastianraschka.com — ~18 hrs
- Alignment: RLHF [arXiv:2203.02155](https://arxiv.org/abs/2203.02155) · DPO [arXiv:2305.18290](https://arxiv.org/abs/2305.18290) — ~10 hrs
- HF LLM course (adaptation units) + Chip Huyen *AI Engineering* — ~14 hrs — [Hugging Face Checkpoint](https://huggingface.co/checkpoint): LoRA/QLoRA mechanics + when to use; RLHF vs DPO; parameter-efficient fine-tuning trade-offs.

Week 3 — Multimodal / VLM (~26 hrs)

- CLIP [arXiv:2103.00020](#) · LLaVA [arXiv:2304.08485](#) · Flamingo [arXiv:2204.14198](#) — **~18 hrs** (read + run a small CLIP retrieval)
- Relevance to your work: multimodal for imaging+text and molecular-graph+text — **~8 hrs** *Checkpoint*: how CLIP aligns modalities; where a VLM helps (and its failure modes) in scientific data.

Week 4-5 — Agents, MCP & evaluation (~52 hrs) — *reframed to scientific/chemistry agents; mirrors your project*

- HF Agents course + HF MCP course — **~18 hrs** — [Hugging Face](#) · [Hugging Face](#)
- Anthropic — Building Effective Agents (patterns: routing, orchestrator-worker, evaluator-optimizer = the **critic loop** in your project) — **~8 hrs** — [Anthropic](#)
- LangGraph (stateful agent graphs) + Andrew Ng short courses — **~12 hrs** — [langchain-ai.github.io](#) · [DeepLearning.AI](#)
- **Agent/critique evaluation** — how to evaluate an agent and a critic honestly (task success, faithfulness, self-consistency) — **~14 hrs** — (Eugene Yan evals [eugeneyan.com](#))

CAPSTONE 4 · public data: an **MCP-based scientific assistant** — a multi-step agent (RAG + tools + memory + a **critic/verifier loop** + eval) over open literature, exposing a molecular-property/UQ tool (ties to 5E/5F). Sources: PMC-OA [ncbi.nlm.nih.gov](#) · arXiv [arxiv.org](#) · PubMed E-utilities [ncbi.nlm.nih.gov](#) · Semantic Scholar API [semanticsscholar.org](#). *Never ingest internal data.* **Phase 6 self-test:** design an agent with a critic/verifier loop and explain how you'd evaluate the critic; MCP tool-exposure end-to-end; RLHF vs DPO; when an agent beats a single well-prompted call (and when it doesn't).

PHASE 7 — Computer Vision Expert (classical / geometric / 3D / edge) · ~175 hrs (+ Capstone 5) · *late Nov 2027 - early Jan 2028*

Goal: round out into a broad CV expert so no vision question catches you flat (video already covered in Phase 2V), and gain real deployment/edge skill. By end you can reason about multi-view geometry, port a model to an optimized C++/TensorRT pipeline, and speak to 3D/neural rendering.

Week 1-2 — Classical & geometric CV (~45 hrs)

- **First Principles of Computer Vision (Nayar)** — **~18 hrs** — [fpcv.cs.columbia.edu](#) · Coursera [Coursera](#)
- **Cyrill Stachniss** (photogrammetry/SLAM lectures) — **~12 hrs** — [YouTube](#)
- **Szeliski (free book)** as reference; **COLMAP / OpenMVG / ORB-SLAM3 / Open3D** hands-on — **~15 hrs** — [szeliski.org](#) · [colmap.github.io](#) · [github/openMVG](#) · [github/ORB_SLAM3](#) · [open3d.org](#) *Whiteboard*: pinhole model + epipolar constraint; SIFT & scale/rotation invariance; SfM pipeline.

Week 3 — Modern DL vision breadth (~35 hrs)

- **Michigan EECS 498 (Justin Johnson)** — **~20 hrs** — [web.eecs.umich.edu](#) · playlist [YouTube](#)
- Detection frameworks — Detectron2 [github/detectron2](#) · MMDetection [github/mmdetection](#) · Ultralytics YOLO [docs.ultralytics.com](#) — **~15 hrs** *Whiteboard*: two-stage vs one-stage detectors; anchor-based vs anchor-free; NMS.

Week 4 — Deployment & optimization for hardware/edge (~50 hrs)

- OpenCV [docs.opencv.org](#) · PyImageSearch [pyimagesearch.com](#) — **~12 hrs**
- **ONNX + TensorRT** (export → optimize → quantize) — **~20 hrs** — [onnx.ai](#) · [developer.nvidia.com](#)
- **CUDA + PMPP (Kirk & Hwu)** + NVIDIA DLI; Triton / DeepStream — **~18 hrs** — [docs.nvidia.com](#) · [nvidia.com](#) · [github/server](#) · [developer.nvidia.com](#) *Whiteboard*: what TensorRT does; INT8 calibration; porting a PyTorch CV model to a real-time C++ pipeline.

Week 5 — 3D vision & neural rendering (~30 hrs) + Capstone 5

- **NeRF** [arXiv:2003.08934](#) · **3D Gaussian Splatting** [arXiv:2308.04079](#) · nerfstudio [docs.nerf.studio](#) · Open3D point clouds [open3d.org](#) — **~30 hrs**

CAPSTONE 5 (deployment) · public data: SfM/calibration/visual-odometry on public sets, **or** take Capstone-1/2's model → optimized C++ inference (TensorRT/ONNX, quantized) with a latency/accuracy report. Datasets: KITTI [cvlibs.net](#) · TUM RGB-D [cvg.cit.tum.de](#) · ETH3D [eth3d.net](#) **Phase 7 self-test:** pinhole + epipolar; SIFT invariance; two-stage vs one-stage; what TensorRT does; port a PyTorch CV model to a real-time C++ pipeline.

PHASE 8 — Research + Big-Tech Interview Crescendo · ~160 hrs · *Jan - mid-Feb 2028* · → Tier 3

Goal: convert the whole plan into offers. RE loop = 2 medium-hard coding + ML fundamentals + ML system design + behavioral (+ a research discussion for research roles). Research loops **ban AI tools** — rehearse unaided.

Week 1-2 — DSA at RE weight (~55 hrs)

- Timed **NeetCode 150 → Blind 75 → company-tagged** — **~55 hrs** — [neetcode.io](#) · [leetcode.com](#) · DesignGurus [designgurus.io](#) · Striver [takeforward.org](#). Blank editor, ~35 min, code must run; ~4-6/day. *Checkpoint*: a random LeetCode hard in ~45 min, clean running code.

Week 3 — Implement-by-hand ML (~35 hrs) · highest value

- Timed, unaided, blank file: attention, a loss (your domain loss), a sampler, a diffusion step, a training loop, backprop, **a GNN layer** — plus an ML-math/theory sweep (gradients, KL, sampling, convergence, "L1 sparsity via gradients"). Draws on Thread M all year. *Checkpoint*: implement attention + a loss + a sampler + a GNN layer unaided, timed.

Week 4 — ML system design (~35 hrs)

- ML system design first (Chip Huyen); then generic system design for loops that include it — ByteByteGo [bytebytego.com](#) · Gaurav Sen [YouTube](#) · DDIA + MIT 6.824 [pdos.csail.mit.edu](#). 8-10 prompts, recorded, self-critiqued — including **a molecular-property serving system, a virtual-screening pipeline, and an imaging pipeline with domain-shift handling**. *Checkpoint*: a full ML-system-design whiteboard in 45 min.

Week 5 — Research job-talk + behavioral (~35 hrs)

- A 20-min talk + Q&A on your **two strongest capstones** (the imaging flagship + the molecular/physics capstone); ~6-8 STAR stories per org; low-level/OO design where the loop includes it; live mocks (Pramp / interviewing.io). **Phase 8 self-test:** LeetCode hard in ~45 min clean; implement attention + a loss + a sampler + a GNN layer unaided/timed; a 45-min system-design whiteboard; a 20-min job-talk; four STAR stories from memory. → **Tier 3 top-lab loops. Intensive interviewing.**

PHASE Q — Quantum Computing + Quantum Machine Learning (full-depth track) · ~305 hrs (+ Capstone Q ~30) · off the job-critical path; concentrated after Phase 8, ~Feb-Apr 2028

Goal: match hands-on VQE/ADAPT-VQE/DMET-VQE work with rigorous theory — derive VQE from the variational principle, explain barren plateaus mathematically, defend ansatz/measurement choices, and discuss QML's real limits honestly (Schuld). Depth on top of experience, off the job-critical path so it never delays the ladder; it also **grounds 5D's physics** (why DFT/FEP are accurate, the classical↔quantum interface).

COI: public molecules only (H₂/LiH/H₂O or a QM9 subset); avoid your Lilly molecules (no keto-enol/HATU specifics) and anything from the vendor engagement.

Intuition-first entry (before the rigorous courses): Quantum Country (spaced-repetition primer) quantum.country · MIT 8.04 (Adams) [MIT OCW](https://ocw.mit.edu/courses/8-04-quantum-physics-i/) — ~10 hrs

Q.1 — Group theory & symmetry (~22 hrs)

- Group theory for physicists (lectures) — ~14 hrs — [YouTube](https://www.youtube.com/watch?v=8v33333333)
- GDL lectures for the Lie-group/equivariance link (shared with 5A) — ~8 hrs — geometricdeeplearning.com *Checkpoint:* groups/representations; why symmetry underlies both quantum operators and equivariant nets.

Q.2 — Quantum-mechanics math (Hilbert spaces, operators, tensor products) (~25 hrs)

- MIT 8.04 (L6-12) — ~15 hrs — [MIT OCW](https://ocw.mit.edu/courses/8-04-quantum-physics-i/) · Quantum Country quantum.country — ~6 hrs · (optional) Schuller "Geometric Anatomy" sampled — ~4 hrs [YouTube](https://www.youtube.com/watch?v=8v33333333) *Whiteboard:* define a Hilbert space; tensor-product two qubit states; unitarity, Hermitian operators, why measurement outcomes are eigenvalues; pure vs mixed states + the density matrix.

Q.3 — Quantum computing fundamentals: the Watrous / IBM track (~80 hrs) · rigorous core

- **Basics of Quantum Information** — ~25 hrs — learning.quantum.ibm.com · YouTube [YouTube](https://www.youtube.com/watch?v=8v33333333)
- **Fundamentals of Quantum Algorithms** — ~25 hrs — learning.quantum.ibm.com
- **General Formulation** (density matrices, channels, POVMs — matters for noise/error mitigation) — ~18 hrs — learning.quantum.ibm.com
- **Foundations of Quantum Error Correction** — ~12 hrs — learning.quantum.ibm.com
- Companion: Nielsen & Chuang ch 1-4, 8, 10 (read alongside). *Whiteboard:* single/multi-qubit gates; entanglement; the standard algorithms (Deutsch-Jozsa → phase estimation) and what each teaches.

Q.4 — Variational algorithms (VQE/ADAPT/DMET/QAOA) (~45 hrs)

- PennyLane chemistry demos (run the derivations behind code you've used) — ~15 hrs — pennylane.ai · PennyLane Codebook (VQAs) pennylane.ai
- VQE (Peruzzo 2014) [Nature](https://arxiv.org/abs/1411.4028) · ADAPT-VQE (Grimsley 2019) [Nature](https://arxiv.org/abs/1709.03489) · DMET-VQE survey [arXiv:2108.08987](https://arxiv.org/abs/2108.08987) — ~18 hrs
- QAOA (Farhi 2014) [arXiv:1411.4028](https://arxiv.org/abs/1411.4028) · Hadfield review [arXiv:1709.03489](https://arxiv.org/abs/1709.03489) · Pauli grouping / measurement (shot-budget) [arXiv:1907.13117](https://arxiv.org/abs/1907.13117) — ~12 hrs *Whiteboard:* derive the variational principle; VQE end-to-end (ansatz, parameter-shift rule, optimizer); why ADAPT-VQE is parameter-efficient; when DMET fragmentation is worth the overhead.

Q.5 — QML theory (the honest-take material) (~42 hrs)

- PennyLane QML topic page (Schuld-curated) — ~10 hrs — pennylane.ai
- Schuld "Taking stock of QML: a critical perspective" (the talk every senior should watch) — ~6 hrs — [YouTube](https://www.youtube.com/watch?v=8v33333333)
- Schuld & Petruccione ch 4-7 · Key papers: QML=kernels [arXiv:2101.11020](https://arxiv.org/abs/2101.11020) · Barren plateaus [Nature](https://arxiv.org/abs/1905.10876) · Expressibility [arXiv:1905.10876](https://arxiv.org/abs/1905.10876) · "Is quantum advantage the right goal?" [arXiv:2203.01340](https://arxiv.org/abs/2203.01340) · PQC encodings [arXiv:2008.08605](https://arxiv.org/abs/2008.08605) — ~26 hrs · courses to pull from: Péré [github/QML-Course](https://github.com/QML-Course) · Hjorth-Jensen [github/QuantumComputingMachineLearning](https://github.com/QuantumComputingMachineLearning) *Whiteboard:* barren plateaus mathematically + mitigations; QML-as-kernel-methods; the honest hype-vs-promise take (cite Schuld).

Q.6 — Quantum chemistry on quantum computers (~60 hrs)

- Qiskit Global Summer School (chemistry + variational tracks) — ~20 hrs — [YouTube](https://www.youtube.com/watch?v=8v33333333)
- IBM Quantum Learning + Sample-based/Krylov Quantum Diagonalization (SQD) — ~25 hrs — learning.quantum.ibm.com · ibm.com
- Build/replicate small end-to-end chemistry pipelines (public molecules) — ~15 hrs

CAPSTONE Q (~30 hrs) · public molecules: H₂/LiH/H₂O or a QM9 subset — vanilla VQE + UCCSD, ADAPT-VQE, DMET-VQE, benchmarked vs classical CCSD/6-31G; 4-page write-up defending ansatz/optimizer/measurement grouping/noise mitigation/basis set + a barren-plateau analysis; optional delta-ML surrogate connecting to the 5A/5D capstones. **Phase Q self-test:** derive VQE from the variational principle; the parameter-shift rule and why it works; a barren-plateau argument; VQE vs QAOA — when each and when no quantum algorithm; honest QML take (Schuld); DMET fragmentation and the classical-quantum interface. → **QML career branch opens.**

14. RESOURCES — unified index (fused: core + all additions, organized by topic)

One list per topic; earlier "v10-core / v11-additions" split removed. Everything a phase needs is together.

Math & foundations (P0/P1 + threaded math) — StatQuest [YouTube](#) · [statquest.org](#) · 3B1B LA [YouTube](#) · 3B1B Calc [YouTube](#) · 3B1B NN [YouTube](#) · MIT 18.06 [MIT OCW](#) · MIT 18.065 [MIT OCW](#) · Stat 110 [projects.iq.harvard.edu](#) · MIT RES.6-012 [MIT OCW](#) · Blitzstein & Hwang [probabilitybook.net](#) · Matrix Cookbook [math.uwaterloo.ca](#) · Matrix Calculus (Parr & Howard) [explained.ai](#) · Karpathy Z2H [YouTube](#) · repo [github/nn-zero-to-hero](#) · Boyd EE364A [web.stanford.edu](#) · Ruder [ruder.io](#) · MacKay ITILA [inference.org.uk](#) · Group theory [YouTube](#) · GDL [geometricdeeplearning.com](#) · Imperial LA [Coursera](#) · Imperial Multivariate Calc [Coursera](#)

Biomedical imaging — core + expansion (P2) — MRI Q&A [mriquestions.com](#) · OHBM [YouTube](#) · Callaghan [YouTube](#) · iBiology [YouTube](#) · AI for Medical Diagnosis [Coursera](#) · DigitalSreeni [YouTube](#) · code [github/python_for_microscopists](#) · CS231n [cs231n.github.io](#) · playlist [YouTube](#) · Conv arithmetic [arXiv:1603.07285](#) · [github/conv_arithmetic](#) · MONAI [github/tutorials](#) · [YouTube](#) · nnU-Net [github/nnUNet](#) · Kitware [kitware.com](#) · TorchIO [torchio.readthedocs.io](#) · Stanford AIMI [aimi.stanford.edu](#) · Metrics Reloaded [Nature](#) · Yarin Gal uncertainty [YouTube](#) · **Papers:** U-Net [arXiv:1505.04597](#) · V-Net [arXiv:1606.04797](#) · Attention U-Net [arXiv:1804.03999](#) · nnU-Net [Nature](#) · Loss survey [arXiv:2006.14822](#) · Generalized Dice [arXiv:1707.03237](#) · Focal Tversky [arXiv:1810.07842](#) · Boundary [arXiv:1812.07032](#) · ViT [arXiv:2010.11929](#) · TransUNet [arXiv:2102.04306](#) · Swin-UNet [arXiv:2105.05537](#) · UNETR [arXiv:2103.10504](#) · Swin UNETR [arXiv:2201.01266](#) · SAM [arXiv:2304.02643](#) · MedSAM [Nature](#) · MedNeXt [arXiv:2303.09975](#) · **Expansion tools:** Cellpose [cellpose.org](#) · StarDist [github/stardist](#) · CellProfiler [cellprofiler.org](#) · Allen Brain Atlas [atlas.brain-map.org](#) · BrainGlobe [brainglobe.info](#) · ANTsPy [github/ANTsPy](#) · SimpleITK [simpleitk.org](#) · OpenSlide [openslide.org](#) · TIAToolbox [github/tiatoolbox](#) · CLAM [github/CLAM](#) · QuPath [qupath.github.io](#) · napari [napari.org](#) · torchstain [github/torchstain](#) · modAL [github/modAL](#) · DINOv2 [github/dinov2](#) · MAE [arXiv:2111.06377](#) · UNI [github/UNI](#) · Virchow (HF) [Hugging Face](#) · **Datasets:** Medical Decathlon [medicaldecathlon.com](#) · BraTS [synapse.org](#) · TCIA [cancerimagingarchive.net](#) · IXI [brain-development.org](#) · LIVECell [github/LIVECell](#) · Sartorius [kaggle.com](#) · BBBC [bbc.broadinstitute.org](#) · JUMP-CP [jump-cellpainting.broadinstitute.org](#) · RxRx [rxrx.ai](#)

Video / temporal (P2V) — VideoMAE [arXiv:2203.12602](#) · TimeSformer [arXiv:2102.05095](#) · ViViT [arXiv:2103.15691](#) · SlowFast [arXiv:1812.03982](#) · SAM 2 [arXiv:2408.00714](#) · [github/sam2](#) · RAFT [arXiv:2003.12039](#) · ByteTrack [arXiv:2110.06864](#) · OC-SORT [arXiv:2203.14360](#) · BoT-SORT [arXiv:2206.14651](#) · HOTA [arXiv:2009.07736](#) · MOTChallenge [motchallenge.net](#) · DAVIS [davischallenge.org](#) · YouTube-VOS [youtube-vos.org](#) · DeepLabCut [deeplabcut.org](#) · SLEAP [sleap.ai](#) · TrackMate [imagej.net](#) · Ultrack [github/ultrack](#) · Cell Tracking Challenge [celltrackingchallenge.net](#) · CalMS21 [data.caltech.edu](#) · MABe [aicrowd.com](#)

DL theory & architectures (P3) — CS25 [web.stanford.edu](#) · [YouTube](#) · CS236 [YouTube](#) · Lilian Weng [lilianweng.github.io](#) · Annotated Transformer [nlp.seas.harvard.edu](#) · HF CV [Hugging Face](#) · HF Diffusion [Hugging Face](#) · RoPE [arXiv:2104.09864](#) · RMSNorm [arXiv:1910.07467](#) · FlashAttention [arXiv:2205.14135](#) · GQA [arXiv:2305.13245](#) · SwiGLU [arXiv:2002.05202](#) · MoE/Switch [arXiv:2101.03961](#) · ConvNeXt [arXiv:2201.03545](#) · DINOv2 [arXiv:2304.07193](#) · DDPM [arXiv:2006.11239](#) · Yang Song score-based [yang-song.net](#) · Lilian Weng diffusion [lilianweng.github.io](#)

Production / MLOps / LLM systems (P4) — CS336 [stanford-cs336.github.io](#) · [YouTube](#) · HF NLP [Hugging Face](#) · CS329S [stanford-cs329s.github.io](#) · Made With ML [madewithml.com](#) · Full Stack DL [fullstackdeeplearning.com](#) · Chip Huyen [huyenchip.com](#) · Hamel Husain [oreilly.com](#) · Eugene Yan [eugeneyan.com](#) · evals [eugeneyan.com](#) · Anthropic contextual retrieval [Anthropic](#) · Constitutional AI [Anthropic](#) · LlamaIndex [docs.llamaindex.ai](#) · RAGAS [docs.ragas.io](#) · DSPy [dspy.ai](#) · Raschka [magazine.sebastianraschka.com](#) · HF multi-GPU [Hugging Face](#) · vLLM [arXiv:2309.06180](#) · ZeRo [arXiv:1910.02054](#)

Molecular ML & cheminformatics (Track 5A-5F) — CS224W [web.stanford.edu](#) · playlist [YouTube](#) · Bronstein GDL [geometricdeeplearning.com](#) · proto-book [arXiv:2104.13478](#) · PyG [pytorch-geometric.readthedocs.io](#) · colabs [pytorch-geometric.readthedocs.io](#) · DGL [dgl.ai](#) · DGL-LifeSci [github/dgl-lifesci](#) · Chemprop [github/chemprop](#) · DeepChem [deepchem.io](#) · RDKit [rdkit.org](#) · Graphormer [arXiv:2106.05234](#) · GraphGPS [arXiv:2205.12454](#) · SchNet [arXiv:1706.08566](#) · DimeNet++ [arXiv:2011.14115](#) · GemNet [arXiv:2106.08903](#) · EGNN [arXiv:2102.09844](#) · e3nn [e3nn.org](#) · tutorial [blondegeek.github.io](#) · NequIP [Nature](#) · MACE [arXiv:2206.07697](#) · Allegro [github/allegro](#) · Equiformer [arXiv:2206.11990](#) · MolFormer [github/molformer](#) · MolCLR [github/MolCLR](#) · ChemBERTa [arXiv:2010.09885](#) · Uni-Mol [github/Uni-Mol](#) · **Generative:** GFlowNet [arXiv:2106.04399](#) · foundations [arXiv:2111.09266](#) · code [github/gflownet](#) · EDM [github/e3_diffusion_for_molecules](#) · GeoDiff [github/GeoDiff](#) · Torsional Diffusion [github/torsional-diffusion](#) · JT-VAE [arXiv:1802.04364](#) · MoFlow [arXiv:2006.10137](#) · MOSES [github/moses](#) · GuacaMol [github/guacamol](#) · **Physics:** GEOM [github/geom](#) · AutoDock Vina [vina.scripps.edu](#) · gnina [github/gnina](#) · DiffDock [github/DiffDock](#) · OpenMM [openmm.org](#) · OpenFE [openfree.energy](#) · MDAnalysis [mdanalysis.org](#) · Making-it-Rain [github/Making-it-rain](#) · alchemistry [alchemistry.org](#) · **UQ:** conformal [arXiv:2107.07511](#) · MAPIE [github/MAPIE](#) · GPyTorch [gpytorch.ai](#) · BoTorch [botorch.org](#) · Murphy PML [probml.github.io](#) · Pyro [pyro.ai](#) · **Applied:** AiZynthFinder [github/aizynthfinder](#) · Open Reaction Database [open-reaction-database.org](#) · BELKA (DEL) [kaggle.com](#) · Flower (FL) [flower.ai](#) · **Benchmarks/datasets:** TDC [tdcommons.ai](#) · MoleculeNet [moleculenet.org](#) · OGB [ogb.stanford.edu](#) · Polaris [polarishub.io](#) · QM9 [quantum-machine.org](#) · PCQM4Mv2 [ogb.stanford.edu](#) · PDBbind [pdbbind-plus.org.cn](#) · DUD-E [dude.docking.org](#) · ChEMBL [ebi.ac.uk](#) · ZINC [zinc.docking.org](#) · **Protein (context):** AlphaFold 2 [Nature](#) · ESM [github/esm](#) · ESM Atlas [esmatlas.com](#) · Boltz [github/boltz](#) · OpenFold [github/openfold](#)

Quantum & QML (Phase Q) — IBM Quantum Learning [learning.quantum.ibm.com](#) · Watrous [YouTube](#) · Basics [learning.quantum.ibm.com](#) · Algorithms [learning.quantum.ibm.com](#) · General Formulation [learning.quantum.ibm.com](#) · Error Correction [learning.quantum.ibm.com](#) · Qiskit [YouTube](#) · QDA/SQD blog [ibm.com](#) · PennyLane codebook [pennylane.ai](#) · PennyLane QML [pennylane.ai](#) · PennyLane chemistry [pennylane.ai](#) · MIT 8.04 [MIT OCW](#) · Quantum Country [quantum.country](#) · Schuller [YouTube](#) · Péré [github/QML-Course](#) · Hjorth-Jensen [github/QuantumComputingMachineLearning](#) · VQE [Nature](#) · ADAPT-VQE [Nature](#) · DMET-VQE [arXiv:2108.08987](#) · QAOA [arXiv:1411.4028](#) · Hadfield [arXiv:1709.03489](#) · Pauli grouping [arXiv:1907.13117](#) · QML=kernels [arXiv:2101.11020](#) · Barren plateaus [Nature](#) · Expressibility [arXiv:1905.10876](#) · Advantage-goal [arXiv:2203.01340](#) · PQC encodings [arXiv:2008.08605](#) · Preskill Ph219 [theory.caltech.edu](#) · TensorFlow Quantum [tensorflow.org](#)

GenAI & Agents (P6) — HF LLM course [Hugging Face](#) · HF Agents [Hugging Face](#) · HF MCP [Hugging Face](#) · Anthropic Effective Agents [Anthropic](#) · DeepLearning.AI [DeepLearning.AI](#) · LangGraph [langchain-ai.github.io](#) · LoRA [arXiv:2106.09685](#) · QLoRA [arXiv:2305.14314](#) · RLHF [arXiv:2203.02155](#) · DPO [arXiv:2305.18290](#) · CLIP [arXiv:2103.00020](#) · LLaVA [arXiv:2304.08485](#) · Flamingo [arXiv:2204.14198](#) · PMC-OA [ncbi.nlm.nih.gov](#) · PubMed E-utilities [ncbi.nlm.nih.gov](#) · Semantic Scholar API [semanticscholar.org](#)

CV expert / 3D (P7) — FPCV [fpcv.cs.columbia.edu](#) · Coursera [Coursera](#) · Stachniss [YouTube](#) · Michigan EECS 498 [web.eecs.umich.edu](#) · playlist [YouTube](#) · Szeliński [szeliski.org](#) · COLMAP [colmap.github.io](#) · OpenMVG [github/openMVG](#) · ORB-SLAM3 [github/ORB_SLAM3](#) · Open3D [open3d.org](#) · Detectron2 [github/detectron2](#) · MMDetection [github/mmdetection](#) · YOLO docs [ultralytics.com](#) · OpenCV docs [opencv.org](#) · PylImageSearch [pyimagesearch.com](#) · ONNX [onnx.ai](#) · TensorRT developer [nvidia.com](#) · CUDA Guide docs [nvidia.com](#) · NVIDIA DLI [nvidia.com](#) · Triton [github/server](#) · DeepStream developer [nvidia.com](#) · NeRF [arXiv:2003.08934](#) · 3DGS [arXiv:2308.04079](#) · nerfstudio [docs.nerf.studio](#) · KITTI [cvlibs.net](#) · TUM RGB-D [cvg.cit.tum.de](#) · ETH3D [eth3d.net](#)

DSA & C++ (Thread T) — learncpp [learncpp.com](#) · The Chernobyl [YouTube](#) · cppreference [en.cppreference.com](#) · CppCon [YouTube](#) · NeetCode [neetcode.io](#) · LeetCode [leetcode.com](#) · MIT 6.006 [MIT OCW](#) · Abdul Bari [YouTube](#) · DesignGurus [designgurus.io](#) · Striver [takeuforward.org](#) · Sean Prashad [seanprashad.com](#) · CP Handbook [cses.fi](#) · CSES [cses.fi](#)

Implement-by-hand (Thread M) — d2l.ai [d2l.ai](#) · labml.ai [nn.labml.ai](#) · Annotated Transformer [nlp.seas.harvard.edu](#) · micrograd

[github/micrograd](#) · [nanoGPT github/nanoGPT](#) · [ML-From-Scratch github/ML-From-Scratch](#) · [minGPT github/minGPT](#) · [tinygrad github/tinygrad](#) · [JAX jax.readthedocs.io](#) · [Equinox docs.kidger.site](#) · [Anki apps.ankiweb.net](#)

Research output (Thread R) — Papers with Code [paperswithcode.com](#) · MICCAI [miccai.org](#) · MIDL [midl.io](#) · MLSB [mlsb.io](#) · LoG [logconference.org](#) · ISCB/ISMB [iscb.org](#) · arXiv [arxiv.org](#) · bioRxiv [biorxiv.org](#) · OpenReview [openreview.net](#) · How to Read a Paper [web.stanford.edu](#) · Connected Papers [connectedpapers.com](#) · Papers We Love [github/papers-we-love](#)

Interview (P8) — NeetCode [neetcode.io](#) · LeetCode [leetcode.com](#) · DesignGurus [designgurus.io](#) · Striver [takeuforward.org](#) · ByteByteGo [bytebytego.com](#) · YouTube · Gaurav Sen [YouTube](#) · MIT 6.824 [pdos.csail.mit.edu](#)

Company ladder — Qure.ai [qure.ai](#) · SigTuple [sigtuple.com](#) · NVIDIA [nvidia.com](#) · GE HealthCare [gehealthcare.com](#) · MSR India [microsoft.com](#) · Google Research India [research.google](#) · Owkin [owkin.com](#) · Recursion [recursion.com](#) · Insitro [insitro.com](#) · Genesis [genesistherapeutics.ai](#) · Iambic [iambic.ai](#) · Chai [chaidiscovery.com](#) · EvolutionaryScale [evolutionaryscale.ai](#) · Profluent [profluent.bio](#) · InstaDeep [instadeep.com](#) · Valence Labs [valencelabs.com](#) · Isomorphic Labs [isomorphiclabs.com](#) · Google DeepMind [deepmind.google](#) · QpiAI [qpiai.tech](#) · IBM Quantum [ibm.com](#) · IISc ERP [iisc.ac.in](#) · EURAXESS [euraxess.ec.europa.eu](#)

A closing note

This detailed edition trades the crispness of the summary v12 for the thing you actually study from: every phase now reads like Phase 5 — week-by-week, each resource carrying its real hour cost and a link, each week ending in something you *make* and something you must be able to *explain unaided*. The shape is deliberate. Hard theory goes in the fresh morning block; building goes in the second; review and reading go in the tired evening; the weekend is a buffer, not a debt. The self-tests, not the calendar, are the gates — if one doesn't pass, the block stretches. Protect sleep and the rest day like production infrastructure, because at 40 hours a week on top of a job they are exactly that. And keep every portfolio artifact public and yours to defend — that single discipline is what turns two years of work into a resume and a set of interviews that hold up under the hardest questioning. Adjust as the squads' priorities move; the public skills here are the durable core beneath whatever the project names happen to be.

15. Appendix — High-value courses (Coursera / edX / other), with effort & sequencing

Do these ADD hours to the plan? Mostly no.

The plan's ~3,000 hours already budget every phase. Each course below is one of two things:

- **⚡ Substitute** — spend a phase's *existing* budgeted hours on this course instead of the free resource → **≈ hour-neutral** (you have Lilly Coursera, so "paid" is free to you).
- **⊕ Supplement** — genuinely *extra*, optional hours — take only for breadth or if your background in that area is thin.

Effort figures are **estimates**: *full* = the whole course/specialization; *audit-select* = the audit-free subset worth watching. Use audit-select unless you want the graded work/certificate.

Default sequencing (which first)

Free plan resource FIRST ▶ course SECOND. The plan is intuition-first and its free resources are curated — lead with them, use a course to *consolidate/broaden*. Go **course-first** only when (a) it's the better structured option (noted per row), (b) the topic is new and you want orientation, or (c) you want a guided spine/accountability. **Never do both in full on a ▲ topic** (the plan's free resources already lead there).

Foundations & math (Phase 0-1)

- **Mathematics for Machine Learning — Imperial** · effort ~70 h full / **~40 h audit** (LA+Calc) · **⚡** · **first:** free (3B1B intuition) ▶ course as the rigorous spine. [Coursera](#)
- **Machine Learning Specialization — Andrew Ng** · ~75 h / **~30 h** · **⚡** · **first:** free (StatQuest is faster); take this only if you want a fuller guided pass. [Coursera](#)
- **MITx Statistics & ML (6.86x etc.)** — **edX** · ~120 h / ~60 h · **⊕** · optional; only if you want graded rigor beyond the plan. [edX](#)

Deep learning core (Phase 1-3)

- **Deep Learning Specialization — Andrew Ng** · ~125 h / **~70 h** · **⚡** · **first:** free (Karpathy for depth) ▶ course for breadth/tuning. [Coursera](#)
- **Practical Deep Learning for Coders — fast.ai (free)** · ~70 h · **⊕** · optional top-down complement if bottom-up theory drags. [fast.ai](#)
- **NLP Specialization — DeepLearning.AI** · ~80 h / ~30 h · **⊕** · optional; ▲ the plan's transformer resources already lead. [Coursera](#)
- **GANs Specialization — DeepLearning.AI** · ~45 h · **⊕** · optional; ▲ the plan's diffusion focus already leads. [Coursera](#)

Biomedical imaging (Phase 2-2V)

- **AI for Medicine Specialization — DeepLearning.AI** · ~55 h / **~20 h** (Diagnosis course) · **⚡** · **first:** free; AI-for-Medical-Diagnosis is already in the plan. [Coursera](#)

Production / MLOps (Phase 4)

- **ML Engineering for Production (MLOps) — Andrew Ng** · ~60 h / **~35 h** · **⚡** · **first:** COURSE (best for the MLOps vocabulary/framework) ▶ then the plan's free hands-on. [Coursera](#)
- **Generative AI with Large Language Models — DeepLearning.AI + AWS** · ~16 h · **⚡** · either; a clean overview to slot into Phase-4 week

Agents & GenAI (Phase 6)

- **DeepLearning.AI Short Courses (free)** · ~2 h each, **~15 h** for several · ⚡ · **first:** COURSE (orientation) ▶ then build; already in the plan. [DeepLearning.AI](#)

Computer vision (Phase 7)

- **First Principles of Computer Vision — Columbia** · ~50 h full / **~18 h audit** · ⚡ · **first:** COURSE — the best structured classical/geometric CV anywhere; already in the plan. [Coursera](#)
- **TensorFlow Developer — DeepLearning.AI** · ~60 h · ☉ · optional; only for TF deployment practice. [Coursera](#)

Molecular ML / drug discovery (Phase 5)

- **Bioinformatics Specialization — UC San Diego** · ~140 h full / **~40 h audit** · ☉ · optional supplement **if your biology is thin**; audit-select. [Coursera](#)
- **Drug-discovery courses — Coursera (browse)** · ~20 h · ☉ · optional domain context for the DMTA cycle. [Coursera](#)
- ▲ For molecular-ML *methods* (GNNs, equivariance, GFlowNets, docking/FEP, conformal UQ) the plan's free DeepChem/PyG/e3nn/RDKit/OpenMM + papers are the real curriculum — no paid course matches this yet.

Quantum / QML (Phase Q)

- **Quantum-computing intro — edX/Coursera** · ~40 h · ☉ · optional; ▲ the plan's free IBM Quantum Learning + PennyLane already lead — skip unless you want a gentler on-ramp. [edX](#)

Net extra hours (the bottom line)

- **Substitutes (⚡): ≈ 0 net extra** — they replace free hours already budgeted (Math-for-ML, DL Spec, AI-for-Medicine, MLOps, GenAI-LLMs, First-Principles-of-CV, the short courses). Swapping in a course changes *what* you watch, not *how many* hours.
- **Supplements (☉): all optional.** The two most worth it for you: **Bioinformatics (~40 h audit, only if biology is thin)** and **fast.ai (~70 h, only if you want a code-first complement)** — everything else is ▲-covered by the free plan. Adding just those two ≈ ~110 h extra, spread across the ~2-year window (or absorbed by the weekend buffer).
- **Rule of thumb:** treat the plan's free resources as the spine, this appendix as hour-neutral **swap-ins (⚡)** plus a few optional **supplements (☉)**. Don't inflate the plan — the ▲ topics are already best-served free.

How to use

1. **Audit first** (free); go graded (or use Lilly Coursera) only for the assignments or a certificate that helps a specific application.
2. **Don't double-spend on ▲ topics.**
3. **Best paid-value picks for you:** Mathematics-for-ML (Imperial), MLOps (Ng), First-Principles-of-CV (Columbia) — and, if biology is thin, UCSD Bioinformatics.